

Cognitive Linguistics

An Introduction

**Vyvyan Evans and
Melanie Green**



Cognitive Linguistics

This book is dedicated to the memory of Larry Trask, 1944–2004, linguist,
scholar, teacher, colleague, mentor and friend.

COGNITIVE LINGUISTICS AN INTRODUCTION

Vyvyan Evans and Melanie Green

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Preface

The nature of this book

This book represents a general introduction to the area of theoretical linguistics known as cognitive linguistics. It consists of three main parts. Part I provides an overview of some of the main aims, assumptions and commitments of the cognitive linguistics enterprise, and provides an indicative sketch of some of the descriptive analyses and theoretical positions that are representative of cognitive linguistics. The next two parts focus on the two best-developed research frameworks in cognitive linguistics: cognitive semantics (Part II), and cognitive approaches to grammar (Part III). Although some cognitive linguists (notably Langacker) have extended their theories to account for phonology as well as meaning and grammar, we will be mainly concerned with meaning and grammar in this book, and will have little to say about phonology. In part, this reflects the fact that phonology has received relatively little attention within cognitive linguistics (although this situation is changing), and in part this reflects our own interests.

Who is this book for?

Our aim has been to provide a reasonably comprehensive general introduction to cognitive linguistics that is accessible enough for undergraduate students at the university level, while also serving as a work of reference both for linguists and for scholars from neighbouring disciplines. While striving for accessibility, we have also retained considerable detail (including relevant citations in the running text), so that readers (including research students and professional linguists unfamiliar with cognitive linguistics, as well as interested readers from

neighbouring disciplines), are provided with a route into the primary literature. In selecting the material presented, and in the presentation itself, we have attempted to provide as balanced a perspective as possible. However, cognitive linguistics represents a collection of approaches rather than a unified theoretical framework, and different authors often take quite distinct positions on similar phenomena, sometimes relying on distinct terminology. It follows that what we present here under the name of ‘cognitive linguistics’ should be understood as a presentation of the cognitive approach ‘as we see it’.

Using the book

We have designed the book so that, in general terms, each chapter builds on preceding chapters. In particular, our decision to present the material on cognitive semantics (Part II) before the material on cognitive approaches to grammar (Part III) reflects the fact that cognitive grammarians assume much of what has been established by cognitive semanticists in developing their approaches. However, because different readers and course tutors will need to use the book in ways tailored to their specific objectives, we have attempted to make Part II and Part III of the book relatively independent so that they can be used for separate courses. The book has sufficient coverage to provide the basis for a number of different courses. We outline below suggestions for ‘routes’ through the book for three different types of course, assuming 12 teaching weeks at the rate of one chapter per week. Of course, these suggestions can be adjusted depending on teaching time available, level of course and so on. The suggestions made here reflect undergraduate courses taught at the University of Sussex, where this textbook was piloted prior to publication.

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March 2005

	Introduction to cognitive linguistics	Cognitive semantics	Cognitive approaches to grammar
Week 1	Ch. 1. What does it mean to know a language?	Ch. 1. What does it mean to know a language?	Ch. 1. What does it mean to know a language?
Week 2	Ch. 2. The nature of cognitive linguistics: assumptions and commitments	Ch. 2. The nature of cognitive linguistics: assumptions and commitments	Ch. 2. The nature of cognitive linguistics: assumptions and commitments
Week 3	Ch. 3. Universals and variation in language, thought and experience	Ch. 5. What is cognitive semantics?	Ch. 14. What is a cognitive approach to grammar?
Week 4	Ch. 4. Language in use: knowledge of language, language change and language acquisition	Ch. 3. Universals and variation in language, thought and experience	Ch. 4. Language in use: knowledge of language, language change and language acquisition
Week 5	Ch 5. What is cognitive semantics?	Ch. 6. Embodiment and conceptual structure	Ch. 15. The conceptual basis of grammar
Week 6	Ch. 6. Embodiment and conceptual structure	Ch. 7. The encyclopaedic view of meaning	Ch. 16. Cognitive grammar: word classes
Week 7	Ch. 7. The encyclopaedic view of meaning	Ch. 8. Categorisation and idealised cognitive models	Ch. 17. Cognitive grammar: constructions
Week 8	Ch. 9. Metaphor and metonymy	Ch. 9. Metaphor and metonymy	Ch. 18. Cognitive grammar: tense, aspect, mood and voice
Week 9	Ch. 14. What is a cognitive approach to grammar?	Ch. 10. Word meaning and radial categories	Ch. 19. Motivating a construction grammar
Week 10	Ch. 19. Motivating a construction grammar	Ch. 11. Meaning construction and mental spaces	Ch. 20. The architecture of construction grammars
Week 11	Ch. 20. The architecture of construction grammars	Ch. 12. Conceptual blending	Ch. 21. Grammaticalisation
Week 12	Ch. 23. Assessing the cognitive linguistics enterprise	Ch. 13. Cognitive semantics in context	Ch. 22. Cognitive approaches to grammar in context



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Secondly, we owe a debt of thanks to our students. Earlier drafts of this textbook were used to teach undergraduate courses at the University of Sussex in ‘Cognitive Semantics’ during the 2003/4 academic year, and ‘Cognitive Grammar’, ‘Grammar and Mind’ and ‘Cognitive Semantics’ during the 2004/5 academic year. These students engaged fully with the text and provided all sorts of practical suggestions that greatly improved both the presentation and the accessibility of the book. We thank them for their patience and friendly criticism. In particular, we would like to acknowledge the following students from the 2003/4 cohort: Alison Barnes, Nicola Buxton, David Cafferty, Hayley Clark, Steven Cuthbert, Laura Daw, Mary Downham, Mark Hall, Esther Hamon, Shinichiro Harigai, Minna Kirjavainen, Andrew Reid, Kristal Robinson, Claire Simmonds, Rebecca Stubbs, Caroline Veale, Daniel Whaley, Hannah Williams and Susannah Williams; and the following from the 2004/5 cohort: Jonathan Archer, Beatrice Ashmore, Thomas Baker, Mathew Baker, James Brown, Joseph Clark, Tegan French, Simon Harber, Sarah Hayward, Simon Knight, Thomas Kuhn, Alice Lee, Alistair Lockie, Melinda Lowing, Tanja Obradovic, Lucy Meyer, Peter Nichols, Sean Palmer, Melanie Pinet,

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Finally, this book is in many ways a product of the unique academic environment at the University of Sussex, most notably the depth and quality of teaching and research in the cognitive sciences, and an atmosphere in which researchers of different theoretical persuasions can share their ideas. It was while the Linguistics and English Language Department was in the School of Cognitive and Computing Sciences (COGS) that the late Professor Larry Trask first proposed and developed undergraduate courses in cognitive linguistics at Sussex, which our students could take alongside courses in formal linguistics and the other cognitive sciences in order to develop a broader and more critical understanding of theoretical models in linguistics. This book emerged from our attempts to teach these courses at a time when appropriate teaching materials were few and far between. Our motivation for writing this book can therefore be directly attributed to the interest in cognitive linguistics, and linguistics in general, that Larry's teaching instilled in others here at Sussex. This book is dedicated to Larry's memory.

Abbreviations, symbols and transcription

A	adjective
ABS	absolute
ACC	accusative
ANT	anterior
AP	adjective phrase
AUX	auxiliary
BEN	benefactive
DEF	definite
ERG	ergative
F	feminine
HPSG	Head-driven Phrase Structure Grammar
ICM	idealised cognitive model
IMPF	imperfective
INF	infinitive
LF	logical form
M	masculine
MOD	modal
N	noun
NEG	negative
NOM	nominative
NP	noun phrase
OBJ	object
P	preposition
PASS	passive
PERF	perfective
PF	phonological form

PL	plural
PP	preposition phrase
P.PART	past participle
PRED	predicate
PRES	present
PROG	progressive
PRT	particle
S	singular
SFOC	focused subject
SUBJ	subject
TR	trajector
LM	landmark
TAM	tense, aspect, mood
V	verb
VP	verb phrase

1	1st person
2	2nd person
3	3rd person

*	ungrammatical
?	marginal
#	unacceptable

/x/	phoneme
[x]	allophone

$\bar{v}$	long vowel
$\dot{v}$	low tone vowel
$\hat{v}$	falling tone vowel

<i>italics</i>	linguistic form
SMALL CAPS	concept

bold bold is used for the first mention of key terms introduced in each chapter and for those terms in the summary at the end of each chapter

Part I: Overview of the cognitive linguistics enterprise



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Introduction

Cognitive linguistics is a modern school of linguistic thought that originally emerged in the early 1970s out of dissatisfaction with formal approaches to language. Cognitive linguistics is also firmly rooted in the emergence of modern cognitive science in the 1960s and 1970s, particularly in work relating to human categorisation, and in earlier traditions such as Gestalt psychology. Early research was dominated in the 1970s and 1980s by a relatively small number of scholars. By the early 1990s, there was a growing proliferation of research in this area, and of researchers who identified themselves as ‘cognitive linguists’. In 1989/90, the International Cognitive Linguistics Society was established, together with the journal *Cognitive Linguistics*. In the words of the eminent cognitive linguist Ronald Langacker ([1991] 2002: xv), this ‘marked the birth of cognitive linguistics as a broadly grounded, self conscious intellectual movement’.

Cognitive linguistics is described as a ‘movement’ or an ‘enterprise’ because it is not a specific theory. Instead, it is an approach that has adopted a common set of guiding principles, assumptions and perspectives which have led to a diverse range of complementary, overlapping (and sometimes competing) theories. For this reason, Part I of this book is concerned with providing a ‘character sketch’ of the most fundamental assumptions and commitments that characterise the enterprise as we see it.

In order to accomplish this, we map out the cognitive linguistics enterprise from a number of perspectives, beginning with the most general perspective and gradually focusing in on more specific issues and areas. The aim of Part I is to provide a number of distinct but complementary angles from which the nature and character of cognitive linguistics can be understood. We also draw comparisons with Generative Grammar along the way, in order to set the

cognitive approach within a broader context and to identify how it departs from this other well known model of language.

In Chapter 1, we begin by looking at language in general and at linguistics, the scientific study of language. By answering the question ‘What does it mean to know a language?’ from the perspective of cognitive linguistics, we provide an introductory insight into the enterprise. The second chapter is more specific and explicitly examines the two commitments that guide research in cognitive linguistics: the ‘Generalisation Commitment’ and the ‘Cognitive Commitment’. We also consider the notion of embodied cognition, and the philosophical doctrine of experiential realism, both of which are central to the enterprise. We also introduce the two main approaches to the study of language and the mind adopted by cognitive linguists: cognitive semantics and cognitive (approaches to) grammar, which serve as the focus for Part II and Part III of the book, respectively.

Chapter 3 addresses the issue of linguistic universals and cross-linguistic variation. By examining how cognitive linguists approach such issues, we begin to get a feel for how cognitive linguistics works in practice. We explore the idea of linguistic universals from typological, formal and cognitive perspectives, and look in detail at patterns of similarity and variation in human language, illustrating with an investigation of how language and language-users encode and conceptualise the domains of SPACE and TIME. Finally, we address the Sapir-Whorf hypothesis: the idea that language can influence non-linguistic thought, and examine the status of this idea from the perspective of cognitive linguistics.

In Chapter 4 we focus on the usage-based approach adopted by cognitive linguistic theories. In particular, we examine how representative usage-based theories attempt to explain knowledge of language, language change and child language acquisition. Finally, we explore how the emphasis on situated language use and context gives rise to new theories of human language that, for the first time, provide a significant challenge to formal theories of language.

What does it mean to know a language?

Cognitive linguists, like other linguists, study language for its own sake; they attempt to describe and account for its **systematicity**, its **structure**, the **functions** it serves and how these functions are realised by the language system. However, an important reason behind why cognitive linguists study language stems from the assumption that language reflects patterns of thought. Therefore, to study language from this perspective is to study patterns of **conceptualisation**. Language offers a window into cognitive function, providing insights into the nature, structure and organisation of thoughts and ideas. The most important way in which cognitive linguistics differs from other approaches to the study of language, then, is that language is assumed to reflect certain fundamental properties and design features of the human mind. As we will see throughout this book, this assumption has far-reaching implications for the scope, methodology and models developed within the cognitive linguistic enterprise. Not least, an important criterion for judging a model of language is whether the model is psychologically plausible.

Cognitive linguistics is a relatively new school of linguistics, and one of the most innovative and exciting approaches to the study of language and thought that has emerged within the modern field of interdisciplinary study known as cognitive science. In this chapter we will begin to get a feel for the issues and concerns of practising cognitive linguists. We will do so by attempting to answer the following question: what does it mean to know a language? The way we approach the question and the answer we come up with will reveal a lot about the approach, perspective and assumptions of cognitive linguists. Moreover, the view of language that we will finish with is quite different from the view suggested by other linguistic frameworks. As we will see throughout this book, particularly in the comparative chapters at the ends of Part II and Part III, the

answer to the title of this chapter will provide a significant challenge to some of these approaches. The cognitive approach also offers exciting glimpses into hitherto hidden aspects of the human mind, human experience and, consequently, what it is to be human.

1.1 What is language for?

We take language for granted, yet we rely upon it throughout our lives in order to perform a range of functions. Imagine how you would accomplish all the things you might do, even in a single day, without language: buying an item in a shop, providing or requesting information, passing the time of day, expressing an opinion, declaring undying love, agreeing or disagreeing, signalling displeasure or happiness, arguing, insulting someone, and so on. Imagine how other forms of behaviour would be accomplished in the absence of language: rituals like marriage, business meetings, using the Internet, the telephone, and so forth. While we could conceivably accomplish some of these things without language (a marriage ceremony, perhaps?), it is less clear how, in the absence of telepathy, making a telephone call or sending an e-mail could be achieved.

In almost all the situations in which we find ourselves, language allows quick and effective expression, and provides a well developed means of **encoding** and **transmitting** complex and subtle ideas. In fact, these notions of encoding and transmitting turn out to be important, as they relate to two key functions associated with language, the **symbolic function** and the **interactive function**.

1.1.1 The symbolic function of language

One crucial function of language is to express thoughts and ideas. That is, language encodes and externalises our thoughts. The way language does this is by using **symbols**. Symbols are ‘bits of language’. These might be meaningful subparts of words (for example, *dis-* as in *distaste*), whole words (for example, *cat*, *run*, *tomorrow*), or ‘strings’ of words (for example, *He couldn’t write a pop jingle let alone a whole musical*). These symbols consist of **forms**, which may be spoken, written or signed, and meanings with which the forms are conventionally paired. In fact, a symbol is better referred to as a **symbolic assembly**, as it consists of two parts that are conventionally associated (Langacker 1987). In other words, this symbolic assembly is a **form-meaning pairing**.

A form can be a sound, as in [kæt]. (Here, the speech sounds are represented by symbols from the International Phonetic Alphabet.) A form might be the orthographic representation that we see on the written page: *cat*, or a signed gesture in a sign language. A **meaning** is the conventional ideational or semantic content associated with the symbol. A symbolic assembly of form and meaning is represented in Figure 1.1.

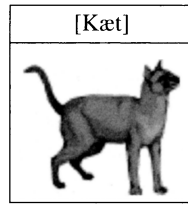


Figure 1.1 A symbolic assembly of form and meaning

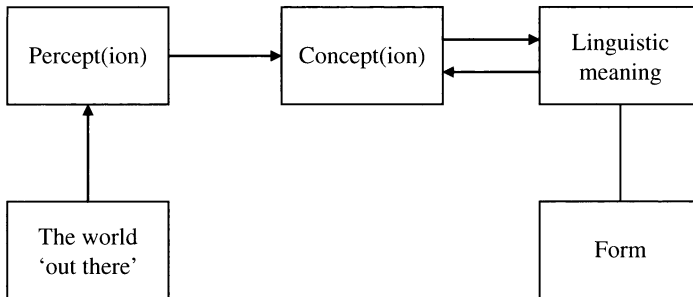


Figure 1.2 Levels of representation

It is important to make it clear that the image of the cat in Figure 1.1 is intended to represent not a particular referent in the world, but the idea of a cat. That is, the image represents the meaning conventionally paired with the form pronounced in English as [kæt]. The meaning associated with a linguistic symbol is linked to a particular mental representation termed a **concept**. Concepts, in turn, derive from **percepts**. For instance, consider a piece of fruit like a pear. Different parts of the brain perceive its shape, colour, texture, taste, smell and so on. This diverse range of perceptual information deriving from the world ‘out there’ is integrated into a single **mental image** (a representation available to consciousness), which gives rise to the concept of PEAR. When we use language and utter the form *pear*, this symbol corresponds to a conventional meaning, and therefore ‘connects’ to a concept rather than directly to a physical object in the external world (see Figure 1.2).

Our cognitive abilities integrate raw perceptual information into a coherent and well defined mental image. The meanings encoded by linguistic symbols then, refer to our **projected reality** (Jackendoff 1983): a mental representation of reality, as construed by the human mind, mediated by our unique perceptual and conceptual systems.

We stated above that the symbolic function of language serves to encode and externalise our thoughts. We are now in a position to qualify this view. While our **conceptualisations** are seemingly unlimited in scope, language represents a limited and indeed limiting system for the expression of thought; we’ve all

experienced the frustration of being unable to ‘put an idea into words’. There is, after all, a finite number of words, with a delimited set of conventional meanings. From this perspective then, language merely provides **prompts** for the construction of a conceptualisation which is far richer and more elaborate than the minimal meanings provided by language (Fauconnier 1997; Turner 1991). Accordingly, what language encodes is not thought in its complex entirety, but instead rudimentary instructions to the conceptual system to access or create rich and elaborate ideas. To illustrate this point, consider the following illustration adapted from Tyler and Evans (2003):

(1) The cat jumped over the wall.

This sentence describes a jump undertaken by a cat. Before reading on, select the diagram in Figure 1.3 that best captures, in your view, the trajectory of the jump.

We anticipate that you selected the fourth diagram, Figure 1.3(d). After all, the conventional interpretation of the sentence is that the cat begins the jump on one side of the wall, moves through an arc-like trajectory, and lands on the other side of the wall. Figure 1.3(d) best captures this interpretation. On first inspection, this exercise seems straightforward. However, even a simple sentence like (1) raises a number of puzzling issues. After all, how do we know that the trajectory of the cat’s jump is of the kind represented in Figure 1.3(d)? What information is there in the sentence that provides this interpretation and excludes the trajectories represented in Figures 1.3(a–c)?

Even though the sentence in (1) would typically be judged as unambiguous, it contains a number of words that have a range of interpretations. The behaviour described by *jump* has the potential to involve a variety of trajectory shapes. For instance, jumping from the ground to the table involves the trajectory represented in Figure 1.3(a). Jumping on a trampoline relates to the trajectory represented in 1.3(b). Bungee jumping involves the trajectory represented in 1.3(c), in which the bungee jumper stops just prior to contact with the surface. Finally, jumping over a puddle, hurdle, wall and so on involves an arc-like trajectory as in 1.3(d).

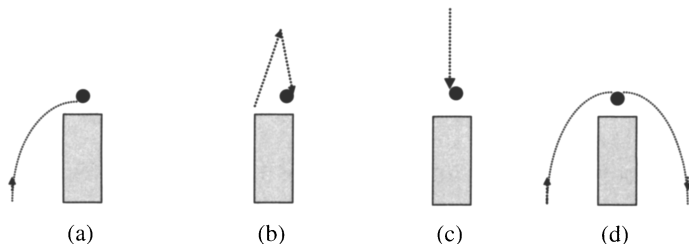


Figure 1.3 Possible trajectories for *The cat jumped over the wall*

If the lexical item *jump* does not in itself specify an arc-like trajectory, but is vague with respect to the shape of the trajectory, then perhaps the preposition *over* is responsible. However, *over* can also have a range of possible interpretations. For instance, it might mean ‘across’, when we walk *over* a bridge (a horizontal trajectory). It might mean ‘above’, when an entity like a hummingbird is *over* a flower (higher than but in close proximity to). Equally, *over* could mean ‘above’ when a plane flies *over* a city (much higher and lacking close proximity). These are just a few of the possibilities. The point to emerge from this brief discussion is that *over* can be used when different kinds or amounts of space are involved, and with a number of different trajectories or paths of motion.

Consider a further complication. Figure 1.3(d) crucially represents the cat’s motion ending at a point on the opposite side of the wall relative to the starting position of the jump. Yet no linguistic element in the sentence explicitly provides us with this information.

Example (1) therefore illustrates the following point: even in a mundane sentence, the words themselves, while providing meanings, are only partially responsible for the conceptualisation that these meanings give rise to. Thought relies on a rich array of encyclopaedic knowledge (Langacker 1987). For example, when constructing an interpretation based on the sentence in (1), this involves at the very least the following knowledge: (1) that the kind of jumping cats perform involves traversing obstacles rather than bungee jumping; (2) that if a cat begins a jump at a point on one side of an obstacle, and passes through a point above that obstacle, then gravity will ensure that the cat comes to rest on the other side of the obstacle; (3) that walls are impenetrable barriers to forward motion; (4) that cats know this, and therefore attempt to circumnavigate the obstacle by going over it. We use all this information (and much more), in constructing the rich conceptualisation associated with the sentence in (1). The words themselves are merely prompts for the construction process.

So far, then, we have established that one of the functions of language is to represent or symbolise concepts. Linguistic symbols, or more precisely symbolic assemblies, enable this by serving as prompts for the construction of much richer conceptualisations. Now let’s turn to the second function of language.

1.1.2 The interactive function of language

In our everyday social encounters, language serves an **interactive function**. It is not sufficient that language merely pairs forms and meanings. These form-meaning pairings must be recognised by, and be accessible to, others in our community. After all, we use language in order to ‘get our ideas across’, in other words to **communicate**. This involves a process of transmission by the speaker, and decoding and interpretation by the hearer, processes that involve the construction of rich conceptualisations (see Figure 1.4).

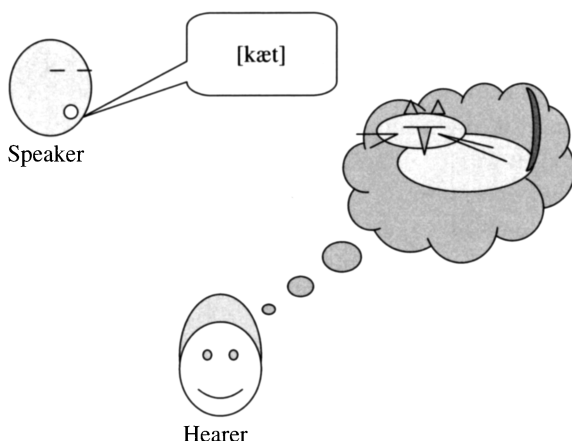


Figure 1.4 The interactive function

The messages we choose to communicate can perform various interactive and **social functions**. For example, we can use language to change the way the world is, or to make things happen:

- (2) a. I now pronounce you man and wife.
- b. Shut the door on your way out!

The utterance in (2a), spoken by a suitably qualified person (such as a member of the clergy licensed to perform marriages), in an appropriate setting (like a church), in the presence of two unmarried adults who consent to be joined in matrimony, has the effect of irrevocably altering the social, legal and even spiritual relationship between the two people. That is, language itself can serve as a **speech act** that forever alters an aspect of our reality.

Similarly, in the example in (2b), the utterance represents a command, which is also a type of speech act. Language provides a means of communication, allowing us to share our wishes and desires. Moreover, the way in which these wishes and desires are expressed signals who we are, and what kind of relationship we have with our addressee. We would be unlikely to issue a command like (2b) to the Queen of England, for example.

Another way in which language fulfils the interactive function relates to the notion of **expressivity**. Language is ‘loaded’, allowing us to express our thoughts and feelings about the world; consider the different mental images evoked by the following expressions, which might be used by different speakers to refer to the same individual:

- (3) a. the eminent linguist
- b. the blonde bombshell

While the example in (3a) focuses on the profession of the individual and her relative standing in that profession, the example in (3b) focuses on her physical appearance. Moreover, although both these sentences relate to a female linguist, the person's gender cannot be inferred from the sentence in (3a) while it can from the second sentence due to normative patterns of linguistic behaviour and social stereotypes. That is, we typically use the expression *blonde bombshell* to describe the physical attributes of women rather than men.

Language also plays a role in how we affect other people in the world, and how we make others feel by our choice of words. That is, language can provide information about **affect** (emotional response):

- (4) a. Shut up!
- b. I'm terribly sorry to interrupt you, but . . .

These examples also illustrate the way in which we present our public selves through language. The language we choose to use conveys information about our attitudes concerning others, ourselves and the situations in which we find ourselves.

Language can be used to create scenes or **frames** of experience, indexing and even constructing a particular context (Fillmore 1982). In other words, language use can invoke frames that summon rich knowledge structures, which serve to call up and fill in background knowledge.

- (5) a. How do you do?
- b. Once upon a time . . .

The example in (5a) creates a greeting frame, signalling an acknowledgement of another person and a recognition that this is the first time they have met. It also signals a degree of formality, which expressions like *hey*, *what's up?* or *hi* would not. Analogously, the utterance in (5b) signals the beginning of a fairy-tale. In other words, just by hearing or reading the expression in (5b) an entire frame is invoked, which guides how we should respond to what follows, what our expectations should be and so forth.

In summary, we've seen that not only does language encode particular meanings, but also that, by virtue of these meanings and the forms employed to symbolise these meanings which constitute part of shared knowledge in a particular speech community, language can serve an interactive function, facilitating and enriching communication in a number of ways.

1.2 The systematic structure of language

Having seen some examples of what language is used for, let's now consider how language is structured. Language is a system for the expression of meaning and

for carrying out its symbolic and interactive functions. So, what evidence is there for the systematicity of language?

1.2.1 Evidence for a system

Language consists of symbolic assemblies that are combined in various ways to perform the functions we described in section 1.1. A symbolic assembly is a conventional **linguistic unit**, which means that it is a piece of language that speakers recognise and ‘agree’ about in terms of what it means and how it is used. As we will see later in the book, particularly in Part III, one of the prominent concerns in cognitive approaches to grammar is how to model the inventory of linguistic units that make up a language. For example, speakers of Modern English ‘agree’ that the form *cat* is used to refer to a certain kind of meaning which we illustrated in Figure 1.2. A conventional unit can be a meaningful sub-part of a word, which linguists call a **morpheme** (*anti-dis-establish . . .*), a whole word, a string of words that ‘belong’ together (a **phrase**) or a whole sentence. Now let’s consider another example:

(6) He kicked the bucket

This utterance consists of a sentence that has an **idiomatic meaning** in English. That is, its meaning is not predictable from the integrated meanings of the individual words. A non-native speaker of English who has not learnt the ‘special’ idiomatic meaning will only be able to interpret example (6) literally. Native speakers of English, on the other hand, while also being able to interpret the sentence literally, often cannot avoid the idiomatic meaning ‘he died’. Of course, whether a literal versus an idiomatic interpretation is accessed depends on the situation or **context** in which the utterance occurs.

Focusing for now on the idiomatic interpretation, we can view this utterance as a unit that has a particular meaning associated with it. Therefore, it counts as a symbolic assembly. Another term for symbolic assembly that is employed by some cognitive linguists is **construction** (e.g. Goldberg 1995). We will look in detail at the notion of symbolic assemblies and constructions in Part III of the book.

When we change certain aspects of the sentence in (6), the meaning is affected. For example, if we change the object (the thing being kicked), as in (7), we lose the idiomatic meaning and are left with a **literal** utterance:

(7) He kicked the mop.

For many cognitive linguists, what makes example (7) ‘literal’ is that this sentence ‘as a whole’ does not represent a construction. Instead, the meaning of (7)

is interpreted by **unifying** the smaller units, the words. In contrast, example (6) is interpreted as a whole single unit: a construction. One way of expressing this idea in more intuitive terms is to use the metaphor of ‘storage’: suppose we store our knowledge of words, phrases and complex constructions in a mental ‘box’. The behaviour of larger constructions, like *kick the bucket*, suggests that these are stored as ‘chunks’ or single units, just like words. The meanings of sentences like (7) on the other hand are ‘built’ by unifying the individual words that make them up.

Now consider another example. If we change the structure of example (6) in the following way, we also lose the idiomatic meaning:

(8) The bucket was kicked by him.

This example shows that, in addition to meaning, constructions (form-meaning pairings) have particular formal grammatical patterns associated with them. In other words, the properties of the construction relate not only to the individual words that make it up, as in (6), but also to the grammatical form, or **word order**. The passive construction in (8), in which *the bucket* is placed in subject position, fails to provide the idiomatic meaning associated with the sentence in (6). We can conclude from this that the linear arrangement of the words in the sentence constitutes part of an individual’s knowledge of idiomatic constructions like (6).

This point is also illustrated by an **ungrammatical** sentence, a sentence that does not correspond to any of the formal patterns associated with the constructions of English, as in (9), and consequently does not have a conventional meaning associated with it. Ungrammaticality is indicated by an asterisk:

(9) *Bucket kicked he the

As we noted above, the sentence in (6) qualifies as a construction because it consists of particular words arranged in a particular order, and these words are conventionally associated with a particular (idiomatic) meaning. However, we have suggested that constructions can also give rise to ‘literal’ meanings. To illustrate this, we will examine another sentence that has both idiomatic and literal meanings. For instance, consider the following linguistic joke:

(10) A: Waiter, what is this fly doing in my soup?
 B: I think that’s the breaststroke, sir!

This joke turns on the ambiguity between the regular interrogative construction, in which a speaker is enquiring after the intention or purpose of something or someone (*What’s that seagull doing on the roof? What’s that woman*

doing over there?), and the ‘What’s X doing Y construction’, studied in detail by cognitive linguists Paul Kay and Charles Fillmore (1999), in which the speaker is indicating that a particular situation is incongruous or unacceptable (*What are you doing wearing those bunny ears? What are those clothes doing on the floor?*). Notice that each of these interpretations requires a different kind of response. For the regular interrogative construction, the response should consist minimally of a piece of information corresponding to the question word (*building a nest; waiting for a bus*). For the ‘what’s X doing Y’ construction, on the other hand, the expected response is typically an explanation, excuse or apology (*I’m going to a fancy-dress party; I’ve been busy*).

Crucially, for example (10), these two very different meanings are conventionally associated with exactly the same words arranged in the same sequence. The humorous effect of the waiter’s reply rests on the fact that he has chosen to respond to the ‘wrong’ interpretation. While the diner is employing the ‘what’s X doing Y’ construction, the waiter prefers to respond to the interrogative construction.

The examples in this section illustrate the fact that there is a systematic relationship between words, their meanings and how they are arranged in conventional patterns. In other words, language has a systematic structure.

1.2.2 The systematic structure of thought

Does the systematic structure found in language reflect a systematic structure within our conceptual system? Cognitive linguists certainly think so. Cognitive linguists explore the hypothesis that certain kinds of linguistic expressions provide evidence that the structure of our conceptual systems is reflected in the patterns of language. Moreover, as we will see throughout this book, the way the mind is structured can be seen as a reflection, in part, of the way the world (including our sociocultural experience) is structured and organised. Consider the examples in (11).

- (11) a. Christmas is fast approaching.
- b. The number of shares we own has gone up.
- c. Those two have a very close friendship.

These examples relate to the abstract **conceptual domains** of TIME (11a), QUANTITY (11b) and AFFECTION (11c). A conceptual domain is a body of knowledge within our conceptual system that contains and organises related ideas and experiences. For example, the conceptual domain of TIME might relate a range of temporal concepts including *Christmas*, which is a temporal event. Notice that in each sentence in (11) the more abstract concepts *Christmas*, *number (of shares)* and *friendship* are understood in terms of conceptual domains relating to *concrete*

physical experience. For instance, Christmas is conceptualised in terms of the domain of physical MOTION, which is evident in the use of the word *approaching* in (11a). Clearly *Christmas* (and other temporal concepts) cannot literally be said to undergo motion. Similarly, the notion of *number of shares* is conceptualised in terms of VERTICAL ELEVATION, which is clear from the use of the phrase *gone up* in (11b). Finally, *friendship* is conceptualised in terms of PHYSICAL PROXIMITY in (11c), which is shown by the use of the word *close*.

One of the major findings to have emerged from studies into the human conceptual system is that abstract concepts are systematically structured in terms of conceptual domains deriving from our experience of the behaviour of physical objects, involving properties like motion, vertical elevation and physical proximity (Lakoff and Johnson 1980, 1999). It seems that the language we use to talk about temporal ideas such as *Christmas* provides powerful evidence that our conceptual system ‘organises’ abstract concepts in terms of more concrete kinds of experiences, which helps to make the abstract concepts more readily accessible.

1.3 What do linguists do?

As we have begun to see, cognitive linguists form hypotheses about the nature of language, and about the conceptual system that it is thought to reflect. These hypotheses are based on observing patterns in the way language is structured and organised. It follows that a theory of language and mind based on linguistic observation must first describe the linguistic facts in a systematic and rigorous manner, and in such a way that the description provides a plausible basis for a speaker’s tacit knowledge of language. This foundation for theorising is termed **descriptive adequacy** (Chomsky 1965; Langacker 1987, 1999a). This concern is one that cognitive linguists share with linguists working in other traditions. Below, we provide an outline of what it is that linguists do and how they go about it.

1.3.1 What?

Linguists try to uncover the systems behind language, to describe these systems and to **model** them. Linguistic models consist of theories about language. Linguists can approach the study of language from various perspectives. Linguists may choose to concentrate on exploring the systems within and between sound, meaning and grammar, or to focus on more applied areas, such as the evolution of language, the acquisition of language by children, language disorders, the questions of how and why language changes over time, or the relationship between language, culture and society. For cognitive linguists, the emphasis is upon relating the systematicity exhibited by language directly to

the way the mind is patterned and structured, and in particular to conceptual structure and organisation. It follows that there is a close relationship between cognitive linguistics and aspects of cognitive psychology. In addition to this, applied linguistics also informs and is informed by the cognitive linguistics research agenda in various ways (see Chapters 3 and 4 for further discussion of this point).

1.3.2 Why?

Linguists are motivated to explore the issues we outlined above by the drive to understand human cognition, or how the human mind works. Language is a uniquely human capacity. Linguistics is therefore one of the **cognitive sciences**, alongside philosophy, psychology, neuroscience and artificial intelligence. Each of these disciplines seeks to explain different (and frequently overlapping) aspects of human cognition. In particular, as we have begun to see, cognitive linguists view language as a system that directly reflects conceptual organisation.

1.3.3 How?

As linguists, we rely upon what language tells us about itself. In other words, it is ordinary language, spoken every day by ordinary people, that makes up the ‘**raw data**’ that linguists use to build their theories. Linguists describe language, and on the basis of its properties, formulate hypotheses about how language is represented in the mind. These hypotheses can be tested in a number of ways.

1.3.4 Speaker intuitions

Native speakers of any given human language will have strong **intuitions** about what combinations of sounds or words are possible in their language, and which interpretations can be paired with which combinations. For example, native speakers of English will agree that example (6), repeated here, is a well-formed sentence, and that it may have two possible meanings:

- (6) He kicked the bucket.

They will also agree that (7) and (8), repeated here, are both well-formed sentences, but that each has only one possible meaning:

- (7) He kicked the mop.
(8) The bucket was kicked by him.

Finally, and perhaps most strikingly, speakers will agree that all of the following examples are impossible in English:

- (12) a. *bucket kicked he the
 b. *kicked bucket the he
 c. *bucket the kicked he
 d. *kicked he bucket the

Facts like these show that language, and speakers' intuitions about language, can be seen as a 'window' to the underlying system. On the basis of the patterns that emerge from the description of language, linguists can begin to build theoretical 'models' of language. A model of language is a set of statements that is designed to capture everything we know about this hidden cognitive system in a way that is principled, based on empirical evidence and psychologically plausible.

1.3.5 Converging evidence

How do cognitive linguists evaluate the adequacy of their models? One way is to consider **converging evidence** (Langacker 1999a). This means that a model must not only explain linguistic knowledge, but must also be consistent with what cognitive scientists know about other areas of cognition, reflecting the view that linguistic structure and organisation are a relatively imprecise but nevertheless indicative reflection of cognitive structure and organisation. By way of illustration, consider the scene in Figure 1.5.

How might we use language to describe a scene like this? Most English speakers will agree that (13a) is an appropriate description but that (13b) is 'odd':

- (13) a. The cat is on the chair.
 b. ?The chair is under the cat.

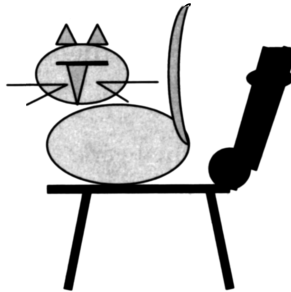


Figure 1.5 *The cat is on the chair*

Why should (13b) be ‘odd’? It’s a perfectly grammatical English sentence. From what psychology has revealed about how the human mind works, we know that we have a tendency to focus our attention on certain aspects of a visual scene. The aspect we focus on is something about which we can make certain predictions. For example, in Figure 1.5 we focus on the cat rather than the chair, because our knowledge of the world tells us that the cat is more likely than the chair to move, to make a noise or to perform some other act. We call this prominent entity the **figure** and the remainder of the scene the **ground**, which is another way of saying ‘background’ (see Chapter 3). Notice that this fact about human psychology provides us with an explanation for why language ‘packages’ information in certain ways. In (13a) *the cat* has a prominent position in the sentence; any theory of language will tell you that sentence initial position is a ‘special’ position in many of the world’s languages. This accords with the prominence of the corresponding entity in the visual scene. This explanation, based on the figure-ground distinction, also provides us with an explanation for why (13b) is ‘odd’. This is an example of how converging evidence works to strengthen or confirm theories of language. Can you think of a situation in which (13b) would not be odd?

1.4 What it means to know a language

Let’s look more closely now at some of the claims made by cognitive linguists about how language is represented in the mind. We have established that the linguist’s task is to uncover the systematicity behind and within language. What kinds of systems might there be within language? We’ll begin to answer this question by introducing one fundamental distinction based on the foundational work of pioneering cognitive linguist Leonard Talmy. Talmy suggests that the **cognitive representation** provided by language can be divided into **lexical** and **grammatical** subsystems. Consider the following example:

(14) **The** hunter tracked **the** tigers.

Notice that certain parts of the sentence in (14) – either whole words (**free morphemes**), or meaningful subparts of words (**bound morphemes**) – have been marked in boldtype. What happens when we alter those parts of the sentence?

- (15) a. **Which** hunter tracked **the** tigers?
b. **The** hunter tracks **the** tigers.
c. **Those** hunters track **a** tiger.

All the sentences in (15) are still about some kind of tracking event involving one or more hunter(s) and one or more tiger(s). What happens when we change

the ‘little’ words like *a*, *the* and *those* and the bound morphemes like *-ed* or *-s* is that we then interpret the event in different ways, relating to information about number (how many hunters or tigers are/were there?), tense (did this event happen before now or is it happening now?), old/new information (does the hearer know which hunters or tigers we’re talking about?) and whether the sentence should be interpreted as a statement or a question.

These linguistic elements and morphemes are known as **closed-class** elements and relate to the grammatical subsystem. The term *closed-class* refers to the fact that it is typically more difficult for a language to add new members to this set of elements. This contrasts with the non-boldtype ‘lexical’ words which are referred to as **open-class**. These relate to the lexical subsystem. The term *open-class* refers to the fact that languages typically find it much easier to add new elements to this subsystem and do so on a regular basis.

In terms of the meaning contributed by each of these two subsystems, while ‘lexical’ words provide ‘rich’ meaning and thus have a **content function**, ‘grammatical’ elements perform a **structuring function** in the sentence. They contribute to the interpretation in important but rather more subtle ways, providing a kind of ‘scaffolding’ which supports and structures the rich content provided by open-class elements. In other words, the elements associated with the grammatical subsystem are constructions that contribute **schematic meaning** rather than rich contentful meaning. This becomes clearer when we alter the other parts of the sentence. Compare (14) with (16):

- (16) a. **The** movie star kissed **the** directors.
b. **The** sunbeam illuminated **the** rooftops.
c. **The** textbook delighted **the** students.

What all the sentences in (16) have in common with (14) is the ‘grammatical’ elements. In other words, the grammatical structure of all the sentences in (16) is identical to that of (15). We know that both participants in the event can easily be identified by the hearer. We know that the event took place before now. We know that there’s only one movie star/sunbeam/textbook, but more than one director/rooftop/student. Notice that the sentences differ in rather a dramatic way, though. They no longer describe the same kind of event at all. This is because the ‘lexical’ elements prompt for certain kinds of concepts that are richer and less schematic in nature than those prompted for by ‘grammatical’ elements. The lexical subsystem relates to things, people, places, events, properties of things and so on. The grammatical subsystem on the other hand relates to concepts having to do with number, time reference, whether a piece of information is old or new, whether the speaker is providing information or requesting information, and so on.

Table 1.1 Properties of the lexical and grammatical subsystems

Lexical subsystem	Grammatical subsystem
Open-class words/morphemes	Closed-class words/morphemes
Content function	Structuring function
Larger set; constantly changing	Smaller set; more resistant to change
Prompts for ‘rich’ concepts, e.g. people, things, places, properties, etc.	Prompts for schematic concepts, e.g. number, time reference, old vs. new, statement vs. question, etc.

A further important distinction between these two subsystems concerns the way that language changes over time. The elements that comprise the lexical (open-class) subsystem make up a large and constantly changing set in any given human language; over a period of time, words that are no longer ‘needed’ disappear and new ones appear. The ‘grammatical’ (closed-class) elements that make up the grammatical subsystem, on the other hand, constitute a smaller set, relatively speaking, and are much more stable. Consequently, they tend to be more resistant to change. However, even ‘grammatical’ elements do change over time. This is a subject we’ll come back to in more detail later in the book when we discuss the process known as **grammaticalisation** (see Chapter 21).

Table 1.1 provides a summary of these important differences between the lexical and grammatical subsystems. Together, these two subsystems allow language to present a cognitive representation, encoding and externalising thoughts and ideas.

Having provided a sketch of what it means to know a language from the perspective of cognitive linguistics, we will now begin to examine the cognitive linguistics enterprise in more detail. In particular, we must consider the assumptions and commitments that underlie the cognitive linguistics enterprise, and begin to examine this approach to language in terms of its perspective, assumptions, the cognitive and linguistic phenomena it considers, its methodologies and its approach to theory construction. We turn to these issues in the next chapter.

1.5 Summary

We began this chapter by stating that cognitive linguists, like other linguists, attempt to describe and account for linguistic **systematicity**, **structure** and **function**. However, for cognitive linguists, language reflects patterns of thought; therefore, to study language is to study patterns of **conceptualisation**. In order to explore these ideas in more detail we looked first at the functions of language. Language provides a means of **encoding** and **transmitting**

ideas: it has a **symbolic function** and an **interactive function**. Language encodes and externalises our thoughts by using **symbols**. Linguistic symbols consist of **form-meaning pairings** termed **symbolic assemblies**. The **meaning** associated with a linguistic symbol relates to a mental representation termed a **concept**. Concepts derive from **percepts**; the range of perceptual information deriving from the world is integrated into a **mental image**. The meanings encoded by linguistic symbols refer to our **projected reality**: a mental representation of reality as construed by the human mind. While our **conceptualisations** are unlimited in scope, language merely provides **prompts** for the construction of conceptualisations. Language also serves an **interactive function**; we use it to **communicate**. Language allows us to perform **speech acts**, or to exhibit **expressivity** and **affect**. Language can also be used to create scenes or contexts; hence, language has the ability to invoke **experiential frames**.

Secondly, we examined the evidence for a linguistic system, introducing the notion of a conventional **linguistic unit**, which may be a **morpheme**, a **word**, a string of words or a sentence. We introduced the notion of **idiomatic meaning** which is available in certain **contexts** and which can be associated with **constructions**. This contrasts with **literal meaning**, which may be derived by **unifying** smaller constructions like individual words. **Word order** constitutes part of an individual's knowledge of particular constructions, a point illustrated by **ungrammatical** sentences. We also related linguistic structure to the systematic structure of thought. **Conceptual domains** reflected in language contain and organise related ideas and experiences.

Next, we outlined the task of the cognitive linguist: to form hypotheses about the nature of language and about the conceptual system that it reflects. These hypotheses must achieve **descriptive adequacy** by describing linguistic facts in a systematic and rigorous manner. Linguists try to uncover, describe and **model** linguistic systems, motivated by the drive to understand human cognition. Linguistics is therefore one of the **cognitive sciences**. Cognitive linguists carry out this task by examining linguistic **data** and by relying on native speaker **intuitions** and **converging evidence**. As an example of converging evidence, we explored the linguistic reflex of the distinction made in psychology between **figure** and **ground**.

Finally, we looked at what it means to know a language, and introduced an important distinction between kinds of linguistic knowledge: the **cognitive representation** provided by language can be divided into **lexical** and **grammatical** subsystems. The lexical subsystem contains **open-class** elements which perform a **content function**. The grammatical subsystem contains **closed-class** elements, which perform a **structuring function** providing **schematic meaning**.

Further reading

A selection of introductory texts that deal broadly with all aspects of linguistics for those relatively new to the subject

- **Dirven and Verspoor (2004)**. This introductory textbook of general linguistics takes a cognitive approach and includes chapters on language and thought, and words, meanings and concepts.
- **Fromkin, Rodman and Hyams (2002)**. A very popular introductory textbook of linguistics.
- **Trask (1999)**. An accessible introduction to linguistics for the layperson; an entertaining read.

A selection of introductory texts on cognitive science in general

- **Bechtel and Graham (eds) (1999)**
- **Cummins and Cummins (eds) (1999)**
- **Green (ed.) (1996)**

Each of these volumes is an introductory-level collection of papers on various aspects of cognitive science. The Green volume places a particular emphasis on linguistics.

A list of texts that provide an overview of the issues of concern to cognitive linguists

- **Allwood and Gärdenfors (eds) (1999)**. A collection of papers on various aspects of cognitive semantics; the paper by Gärdenfors provides a particularly useful overview.
- **Geeraerts (1995)**. This article compares cognitive linguistic approaches with cognitive science and generative grammar and provides a very broad survey of work on cognitive linguistics; not as accessible as Radden's chapter.
- **Geeraerts and Cuyckens (2005)**. An important reference work featuring articles on a wide range of areas in cognitive linguistics by leading scholars in the field.
- **Goldberg (ed.) (1996)**. A collection of conference papers. Provides a representative sample of the range of concerns and issues addressed by cognitive linguists.
- **Janssen and Redeker (1999)**. A collection of papers by some of the leading proponents in the field; a good background to cognitive linguistics in general.

- **Lakoff (1987)**. Seminal text for cognitive linguistics; lively and accessible.
- **Radden (1992)**. Provides a clear and accessible overview of iconicity in language, categorisation, metaphor, cultural models and grammar as a conceptual organising system.
- **Rudzka-Ostyn (1988)**. An early collection. Includes seminal papers by, among others, two highly influential scholars, Langacker and Talmy.

A list of texts that relate to the issues dealt with in this chapter

- **Evans (2004a)**. Explores the relationship between language and conceptual organisation by focusing on how we think and talk about time and temporal experience.
- **Fillmore, Kay and O'Connor (1988)**. Seminal article on the relation between idiomaticity and constructions.
- **Lakoff and Johnson (1980)**. An early but hugely influential study which first proposed that language reflects systematic 'mappings' (conceptual metaphors) between abstract and concrete conceptual domains.
- **Langacker (1999a)**. A survey article which deals with the notions of the symbolic (in Langacker's terms 'semiotic') and interactive functions associated with language, the notion of converging evidence, and how cognitive linguistics differs from formal and functional approaches to language.
- **Nuyts and Pederson (eds) (1997)**. The first chapter provides a good general discussion of the nature of the relationship between language and thought.
- **Talmy (2000)**. Chapter 1 deals with the notion of the cognitive representation and the distinction between the lexical (open-class) and grammatical (closed-class) subsystems.
- **Tyler and Evans (2003)**. The first chapter addresses the idea that words are merely impoverished 'prompts' for rich conceptualisation. Includes a detailed discussion and illustration of the *The cat jumped over the wall* example.

Exercises

1.1 Linguistic encoding

Consider the following examples in the light of our discussion of example (1). Using the diagrams in Figure 1.3 as a starting point, try to draw similar diagrams that capture the path of motion involved in each example. In each case, how

much of this information is explicitly encoded within the meanings of the words themselves? How much seems to depend on what you know about the world?

- (a) The baby threw the rattle out of the buggy.
- (b) I threw the cat out of the back door.
- (c) I tore up the letter and threw it out of the window.
- (d) I threw the tennis ball out of the house.
- (e) I threw the flowers out of the vase.

1.2 Constructions

The examples below contain idiomatic constructions. If you are a non-native speaker of English, you may need to consult a native speaker or a dictionary of idioms to find out the idiomatic meaning. In the light of our discussion of example (6), try changing certain aspects of each sentence to see whether these examples pattern in the same way. For instance, what happens if you change the subject of the sentence (for example, *the presidential candidate* in the first sentence)? What happens if you change the object (for example, *the towel*)? It's not always possible to make a sentence passive, but what happens to the meaning here if you can?

- (a) The presidential candidate threw in the towel.
- (b) Before the exam, Mary got cold feet.
- (c) She's been giving me the cold shoulder lately.
- (d) You are the apple of my eye.
- (e) She's banging her head against a brick wall.

What do your findings suggest about an individual's knowledge of such constructions as opposed to sentences containing literal meaning? Do any of these examples also have a literal meaning?

1.3 Word order

Take example (b) from exercise 1.2 above. Believe it or not, a sentence like this with seven words has 5,040 mathematically possible word order permutations! Try to work out how many of these permutations result in a grammatical sentence. What do your findings suggest?

1.4 Concepts and conceptual domains

The examples below contain linguistic expressions that express abstract concepts. In the light of our discussion of the examples in (11), identify the relevant

conceptual domain that the concept might relate to. Do these abstract concepts appear to be understood in terms of concrete physical experiences? What is the evidence for your conclusions?

- (a) You've just given me a really good idea.
- (b) How much time did you spend on this essay?
- (c) He fell into a deep depression.
- (d) The Stock Market crashed on Black Wednesday.
- (e) Unfortunately, your argument lacks a solid foundation.

Now come up with other sentences which illustrate similar patterns for the following conceptual domains:

- (f) THEORIES
- (g) LOVE
- (h) ARGUMENT
- (i) ANGER
- (j) KNOWING/UNDERSTANDING

1.5 Figure and ground

Consider the scenes in Figure 1.6. For each one, state the sentence that springs first to mind as the most natural way of describing the scene. For example, for the scene in (a), you might come up with *The goldfish is in the bowl*. What happens if you change the sentence around as we did for example (15)? What do your findings suggest about the figure/ground distinction?

1.6 Open-class or closed-class?

Consider the example below in the light of our discussion of examples (15)–(16). First, try to identify the open-class words/morphemes and the closed-class words/morphemes by referring to the properties described in Table 1.1. Next, come up with a set of examples in which only the closed-class words/morphemes have been altered. What kinds of differences do these changes make to the sentence? Finally, try changing the open-class words/morphemes. What kinds of differences do these changes make to the sentence?

The supermodel was putting on her lipstick.

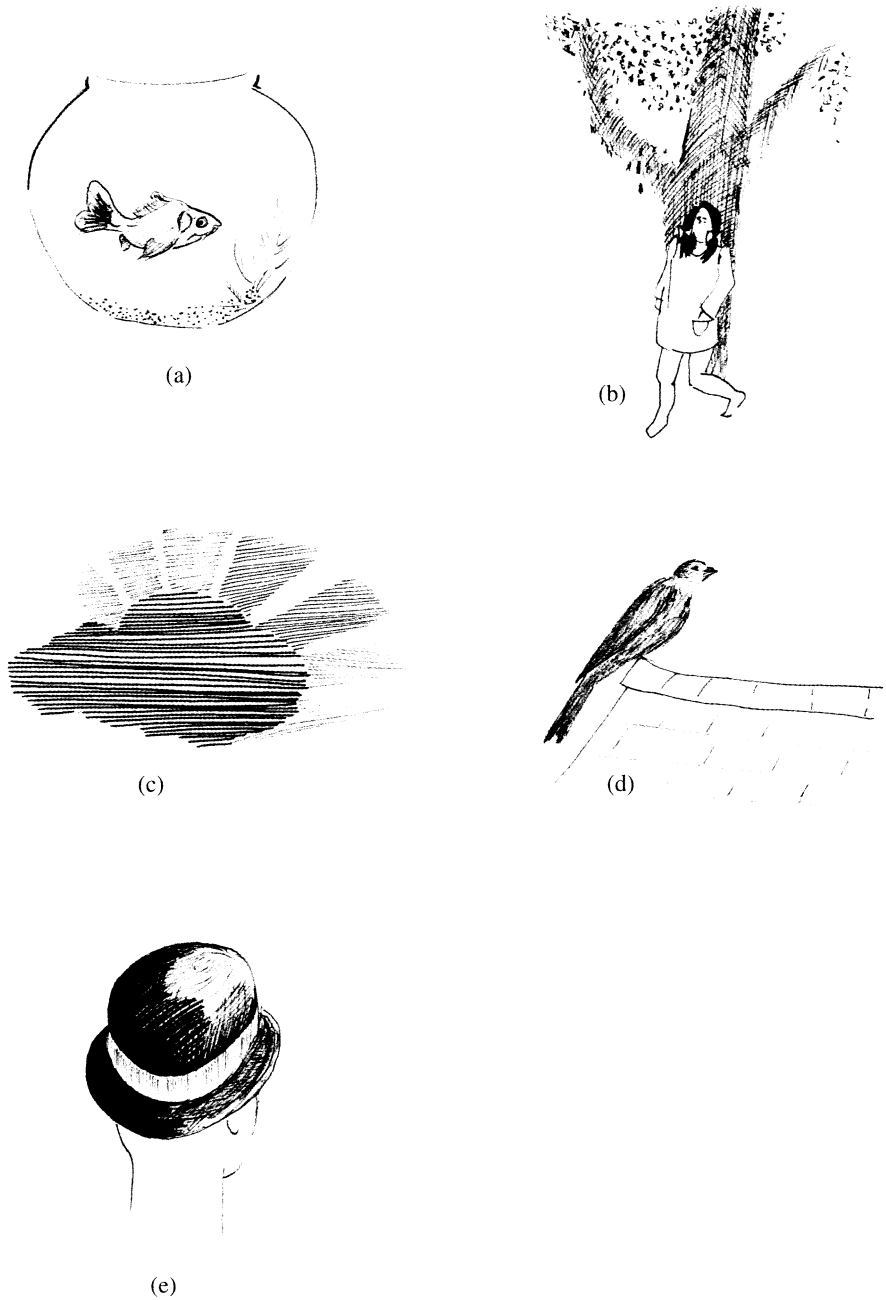


Figure 1.6 Figure and ground

The nature of cognitive linguistics: Assumptions and commitments

In this chapter we address the assumptions and commitments that make cognitive linguistics a distinctive enterprise. We begin by outlining two key commitments widely shared by cognitive linguists. These are the ‘**Generalisation Commitment**’ and the ‘**Cognitive Commitment**’. These two commitments underlie the orientation and approach adopted by practising cognitive linguists, and the assumptions and methodologies employed in the two main branches of the cognitive linguistics enterprise: **cognitive semantics** and **cognitive approaches to grammar**. Once we have outlined the two commitments of cognitive linguistics, we then proceed to address the relationship between language, the mind and experience. The **embodied cognition thesis** is also addressed in some detail as it is at the heart of much research within cognitive linguistics. This thesis holds that the human mind and conceptual organisation are functions of the ways in which our species-specific bodies interact with the environment we inhabit. Finally, we provide a brief overview and introduction to cognitive semantics and cognitive (approaches to) grammar, which are addressed in detail in Parts II and Part III of the book, respectively.

2.1 Two key commitments

In an important 1990 paper, George Lakoff, one of the pioneering figures in cognitive linguistics, argued that the cognitive linguistics enterprise is characterised by two key commitments. These are (1) the ‘Generalisation Commitment’: a commitment to the characterisation of general principles that are responsible for all aspects of human language, and (2) the Cognitive Commitment: a commitment to providing a characterisation of general principles for language that

accords with what is known about the mind and brain from other disciplines. In this section we discuss these two commitments and their implications.

2.1.1 The 'Generalisation Commitment'

One of the assumptions that cognitive linguists make is that there are common structuring principles that hold across different aspects of language, and that an important function of linguistics is to identify these common principles. In modern linguistics, the study of language is often separated into distinct areas such as phonology (sound), semantics (word and sentence meaning), pragmatics (meaning in discourse context), morphology (word structure) syntax (sentence structure) and so on. This is particularly true of **formal approaches**: approaches to modelling language that posit explicit mechanical devices or procedures operating on theoretical **primitives** in order to produce the complete set of linguistic possibilities in a given language. Within formal approaches (such as the Generative Grammar approach developed by Noam Chomsky), it is usually argued that areas such as phonology, semantics and syntax concern significantly different kinds of structuring principles operating over different kinds of primitives. For instance, a syntax 'module' is an area in the mind concerned with structuring words into sentences, whereas a phonology 'module' is concerned with structuring sounds into patterns permitted by the rules of any given language, and by human language in general. This modular view of mind reinforces the idea that modern linguistics is justified in separating the study of language into distinct subdisciplines, not only on grounds of practicality but because the components of language are wholly distinct and, in terms of organisation, incommensurable.

Cognitive linguistics acknowledges that it may often be useful, for practical purposes, to treat areas such as syntax, semantics and phonology as being notionally distinct. The study of syntactic organisation involves, at least in part, the study of slightly different kinds of cognitive and linguistic phenomena than the study of phonological organisation. However, given the 'Generalisation Commitment', cognitive linguists disagree that the 'modules' or 'subsystems' of language are organised in significantly divergent ways, or indeed that distinct modules or subsystems even exist. Below we briefly consider the properties of three areas of language in order to give an idea of how apparently distinct language components can be seen to share fundamental organisational features. The three areas we will look at are (1) categorisation, (2) polysemy and (3) metaphor.

Categorisation

An important recent finding in cognitive psychology is that categorisation is not criterial. This means that it is not an 'all-or-nothing' affair. Instead, human

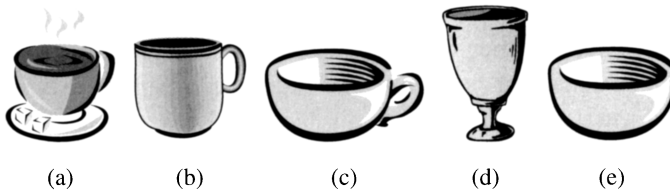


Figure 2.1 Some members of the category CUP

categories often appear to be **fuzzy** in nature, with some members of a category appearing to be more central and others more peripheral. Moreover, degree of **centrality** is often a function of the way we interact with a particular category at any given time. By way of illustration, consider the images in Figure 2.1. It is likely that speakers of English would select the first image 2.1(a) as being more representative of the category CUP than image 2.1(e). However, when drinking from the container in 2.1(e), a speaker might refer to it as *a cup*. On another occasion, perhaps when using a spoon to eat soup from the same container, the same speaker might describe it as *a bowl*. This illustrates that not only is categorisation fuzzy (for example, when does a cup become a bowl?), but also our interaction with a particular entity can influence how we categorise it.

Although the category members in Figure 2.1 may be rated as being more or less **representative** of the category CUP, each of the members appears to resemble others in a variety of ways, despite the fact that there may not be a single way in which all the members resemble each other. For instance, while the cup in 2.1(a) has a handle and a saucer and is used for drinking beverages like tea or coffee, the ‘cup’ in 2.1(d) does not have a handle, nor is it likely to be used for hot beverages like tea or coffee; instead, this cup is more likely to contain drinks like wine. Similarly, while the ‘cup’ in 2.1(e) might be categorised as a ‘bowl’ when we use a spoon to ‘eat’ from it, when we hold the ‘bowl’ to our lips and drink soup from it, we might be more inclined to think of it as a ‘cup’. Hence, although the ‘cups’ in Figure 2.1 vary in terms of how representative they are, they are clearly related to one another. Categories that exhibit degrees of centrality, with some members being more or less like other members of a category rather than sharing a single defining trait, are said to exhibit **family resemblance**.

However, fuzziness and family resemblance are not just features that apply to physical objects like cups; these features apply to linguistic categories like morphemes and words too. Moreover, category-structuring principles of this kind are not restricted to specific kinds of linguistic knowledge but apply across the board. In other words, linguistic categories – whether they relate to phonology, syntax or morphology – all appear to exhibit these phenomena. Formal approaches to linguistics have tended towards the view that a particular category exhibits uniform behaviour which characterises the category. As we will see,

however, linguistic categories, despite being related, often do not behave in a uniform way. Instead, they reveal themselves to contain members that exhibit quite divergent behaviour. In this sense, linguistic categories exhibit fuzziness and family resemblance. We illustrate this below – based on discussion in Taylor (2003) – with one example from each of the following areas: morphology, syntax and phonology.

Categorisation in morphology: the diminutive in Italian

In linguistics, the term ‘diminutive’ refers to an affix added to a word to convey the meaning ‘small’, and is also used to refer to a word formed by the addition of this affix. In Italian the diminutive suffix has a number of forms such as *-ino*, *-etto*, and *-ello*:

- (1) paese → paesino
 ‘village’ ‘small village’

While a common meaning associated with this form is ‘physically small’, as in (1), this is not the only meaning. In the following example the diminutive signals affection rather than small size:

- (2) mamma → mammina
 ‘mum’ ‘mummy’

When applied to abstract nouns, the diminutive acquires a meaning of short temporal duration, reduced strength or reduced scale:

- (3) sinfonia → sinfonia
 ‘symphony’ ‘sinfonia’ (a shorter symphony, often with fewer instruments)

(4) cena → cenetta
 ‘supper’ ‘light supper’

(5) pioggia → ‘pioggerella
 ‘rain’ ‘drizzle’

When the diminutive is suffixed to adjective or adverbs, it serves to reduce intensity or extent:

- (6) bello → bellino
 ‘beautiful’ ‘pretty/cute’

- (7) bene → benino
 ‘well’ ‘quite well’

When the diminutive is added to verbs (the verbal diminutive suffixes are *-icchiare* and *-ucchiare*) a process of intermittent or poor quality is signalled:

- (8) dormire → dormicchiare
 ‘sleep’ ‘snooze’
- (9) lavorare → lavoricchiare
 ‘work’ ‘work half-heartedly’
- (10) parlare → parlucchiare
 ‘speak’ ‘speak badly’ [e.g. a foreign language]

What these examples illustrate is that the diminutive in Italian doesn’t have a single meaning associated with it, but instead constitutes a category of meanings which behave in a variety of distinct ways but nonetheless do appear to be related to one another. The category shares a related form and a related set of meanings: a reduction in size, quantity or quality. Hence, the category exhibits family resemblance.

Categorisation in syntax: ‘parts of speech’

The received view in linguistics is that words can be classified into classes such as ‘noun’ and ‘verb’, traditionally referred to as **parts of speech**. According to this view, words can be classified according to their morphological and distributional behaviour. For example, a word formed by the addition of a suffix like *-ness* (for example, *happi-ness*) is a noun; a word that can take the plural suffix *-s* (for example, *cat-s*) is a noun; and a word that can fill the gap following a sequence of determiner *the* plus adjective *funny* (for example, *the funny ____*) is a noun. In modern linguistics, the existence of word classes is posited not only for practical purposes (that is, to provide us with a tool of description), but also in an attempt to explain how it is that speakers ‘know’ how to build new words and how to combine words into grammatical sentences. In other words, many linguists think that these word classes have psychological reality.

However, when we examine the grammatical behaviour of nouns and verbs, there is often significant variation in the nature of the grammatical ‘rules’ they observe. This suggests that the categories ‘noun’ and ‘verb’ are not homogeneous, but instead that certain nouns and verbs are ‘nounier’ or ‘verbier’ – and hence more representative – than others. In this sense, parts of speech constitute fuzzy categories.

By way of illustration, consider first the **agentive nominalisation** of **transitive** verbs. A transitive verb is a verb that can take an object, such as *import*

(e.g. rugs) and *know* (e.g. a fact). However, while transitive verbs can often be nominalised – that is, made into ‘agentive’ nouns like *driver*, *singer* and *helper* – some verbs, such as *know*, cannot be:

- (11) a. John imports rugs →
John is an importer of rugs
b. John knew that fact →
*John was the knower of that fact

Now consider a second example. While verbs can often be substituted by the ‘be V-able’ construction, this does not always give rise to a well-formed sentence:

- (12) a. His handwriting can be read →
His handwriting is readable
b. The lighthouse can be spotted →
*The lighthouse is spottable

Finally, while most transitive verbs undergo **passivisation**, not all do:

- (13) a. John kicked the ball →
The ball was kicked by John
b. John owes two pounds →
*?Two pounds are owed by John

Despite these differences, these verbs do share some common ‘verbish’ behaviour. For example, they can all take the third person present tense suffix *-s* (*s/he import-s/know-s/read-s/spot-s/kick-s/owe-s* . . .). Therefore, while certain verbs fail to display some aspects of ‘typical’ verb behaviour, this does not mean that these are not part of the category VERB. In contrast, this variation shows us that there is not a fixed set of criteria that serves to define what it means to be a verb. In other words, the linguistic category VERB contains members that are broadly similar yet exhibit variable behaviour, rather like the physical artefact category CUP.

Now let’s consider the linguistic category NOUN. While nouns can be broadly classified according to the morphological and distributional criteria we outlined above, they also show considerable variation. For example, only some nouns can undergo what formal linguists call **double raising**. This term applies to a process whereby a noun phrase ‘moves’ from an embedded clause to the subject position of the main clause via the subject position of another embedded clause. If you are not familiar with the grammatical terms ‘noun phrase’, ‘subject’ or ‘(embedded) clause’, the schematic representation in (14) should help. Noun

phrases, which are units built around nouns (but sometimes consist only of nouns (for example in the case of pronouns like *me* or proper names like *George*), are shown in boldtype. Square brackets represent the embedded clauses (sentences inside sentences) and the arrows show the ‘movement’. Subject positions are underlined:

- (14) a. It is likely [____ to be shown [that **John** has cheated]] →
 b. **John** is likely [____ to be shown [____ to have cheated]]
-

As these examples show, the noun phrase (NP) *John* can only occupy the subject position of a finite or tensed clause: when the verb appears in its ‘to infinitive’ form (for example, *to be/to have*), the NP *John* (which we interpret as the ‘doer’ of the cheating regardless of its position within the sentence) has to ‘move up’ the sentence until it finds a finite verb like *is*. However, some nouns, like *headway*, do not show the same grammatical behaviour:

- (15) a. It is likely [____ to be shown [that **no headway** has been made]]
 →
 b. ***No headway** is likely [____ to be shown [____ to have been made]]
-

Our next example of variation in the behaviour of nouns concerns question tag formation, a process whereby a **tag question** such as *isn’t it?*, *don’t you?* or *mustn’t he?* can be tagged onto a sentence, where it picks up the reference of some previously mentioned unit. For example, in the sentence *Bond loves blondes, doesn’t he?* The pronoun *he* refers back to the subject noun phrase *Bond*. Despite the fact that this grammatical process can apply more or less freely to any subject noun phrase, Taylor (2003: 214) argues that there are nevertheless ‘some dubious cases’. For example, the use of a question tag with the noun *heed* is at best marginal:

- (16) a. Some headway has been made. →
 Some headway has been made, hasn’t it?
 b. Little heed was paid to her. →
 ?*Little heed was paid to her, was it?

As we saw with verbs, examples can always be found that illustrate behaviour that is at odds with the ‘typical’ behaviour of this category. Although most

linguists would not consider this variation sufficient grounds for abandoning the notion of word classes altogether, this variation nevertheless illustrates that categories like NOUN and VERB are not uniform in nature, but are ‘graded’ in the sense that members of these categories exhibit variable behaviour.

Categorisation in phonology: distinctive features

One of the fundamental concepts in phonology is the **distinctive feature**: an articulatory feature that serves to distinguish speech sounds. For example, the sounds /b/ and /p/ are identical in terms of **place** and **manner** of articulation: both are bilabial sounds (produced by bringing the two lips together) and both are plosives (produced by momentary interruption of the airflow followed by sudden release). However, the two sounds are distinguished by the single feature **voice**: the phenomenon whereby the vocal folds in the larynx are drawn tightly together and vibrate as air passes through them, which affects the quality of the sound. The speech sound /b/ is voiced, whereas /p/ is produced with the vocal folds drawn apart, and is therefore unvoiced. This articulatory feature distinguishes many pairs of consonant sounds that otherwise have a similar manner and place of articulation, for example: /t/ and /d/, as in *tug* versus *dug*; /k/ and /g/, as in *curl* versus *girl*; and /s/ and /z/, as in *Sue* versus *zoo*.

In phonology, these **distinctive features** are traditionally viewed as binary features. In other words, a speech sound can be described in terms of whether it has a positive or a negative value for a certain feature. Binary features are popular in formal linguistics, because they enable linguists to describe units of language by means of a set of properties known as a **feature matrix**. This approach has proven particularly successful in phonology. For example, the sounds /p/ and /b/ can be characterised as follows:

$$(17) \quad \begin{array}{c} /p/ \\ \left(\begin{array}{c} + \text{ bilabial} \\ + \text{ plosive} \\ - \text{ voice} \end{array} \right) \end{array} \quad \begin{array}{c} /b/ \\ \left(\begin{array}{c} + \text{ bilabial} \\ + \text{ plosive} \\ + \text{ voice} \end{array} \right) \end{array}$$

However, Jaeger and Ohala (1984) presented research that questions the assumption that distinctive features are binary in nature. In fact, Jaeger and Ohala found that features like voice are judged by actual users of language as graded or fuzzy categories. Jaeger and Ohala trained naive speakers of English (that is, non-linguists), so that they could identify sounds according to whether they were [+ voice] or [– voice]. They then asked subjects to rate the English plosives, fricatives, nasals and semi-vowels in terms of the voice feature. While plosives involve a sudden release of air from the mouth, fricatives are produced by the gradual release of airflow in the mouth: these are sounds like /f/, /v/, /s/, /z/, and so on. Nasals like /m/ and /n/ involve continuous (uninterrupted) airflow

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The researchers found that these sounds were not consistently judged as either voiced or unvoiced. Instead, some sounds were judged as ‘more’ or ‘less’ voiced than others. The ‘voice continuum’ that resulted from Jaeger and Ohala’s study is shown in (18a):

- (18) a. ← most voiced least voiced →
 /r,m,n/ /v,d,z/ /w,j/ /b,d,g/ /f,θ,s,h,f/ /p,t,k/
 b. /r,m,n/ /v,d,z/ /w,j/ /b,d,g/ /f,θ,s,h,f/ /p,t,k/
 ←————— voiced —————→ ←—— voiceless ——→

The sounds were rated accurately by Jaeger and Ohala's subjects in the sense that voiced and voiceless sounds do not overlap but can be partitioned at a single point on this continuum, as shown in (18b). However, what is striking is that the subjects judged some voiced sounds (like /m/) as 'more voiced' than others (like /z/). These findings suggest that the phonological category VOICED SOUNDS also behaves like a fuzzy category.

Taken together, the examples we have considered from the three ‘core’ structural areas of human language – morphology, syntax and phonology – suggest that the nature of the linguistic categories we find in each of these areas can be described in rather similar terms. In other words, at least in terms of categorisation, we can generalise across what are often thought of as wholly distinct kinds of linguistic phenomena.

It is worth pointing out at this stage that cognitive linguistics is not unique in seeking to generalise across these ‘distinct’ areas of human language. Indeed, the quest for binary features in formal linguistics is one example of such an attempt. Encouraged by the relative usefulness of this approach in the area of phonology, formal linguists have, with varying degrees of success, also attempted to characterise word meaning and word classes in terms of binary features. This approach reflects an attempt to capture what are, according to many linguists, the fundamental properties of human language: the ‘design features’ **discreteness** and **duality of patterning**. Broadly, these features refer to the fact that human language is made of smaller discrete units (like speech sounds, morphemes and words) that can be combined into larger units (like morphemes, words and sentences), and that the capacity for varying the patterns of combination is part of what gives human language its infinite creativity (compare *bin* with *nib*, or *Bond loves blondes* with *blondes love Bond*, for example). Thus different theories of human language are often united in pursuing the same ultimate objectives – here, generalisation – but differ in terms of where and how they seek to reach these objectives.

Polysemy

Polysemy is the phenomenon where a single linguistic unit exhibits multiple distinct yet related meanings. Traditionally, this term is restricted to the area of word meaning (**lexical semantics**), where it is used to describe words like *body* which has a range of distinct meanings that are nevertheless related (for example, the human body; a corpse; the trunk of the human body; the main or central part of something). Polysemy is contrasted with **homonymy**, where two words are pronounced and/or spelt the same way, but have distinct meanings (compare *sole* with *soul*, for example, which are pronounced the same way but which no speaker of English would be likely to judge as having related meanings).

Cognitive linguists argue that polysemy is not restricted to word meaning but is a fundamental feature of human language. According to this view, the ‘distinct’ areas of language all exhibit polysemy. Cognitive linguists therefore view polysemy as a key to generalisation across a range of ‘distinct’ phenomena, and argue that polysemy reveals important fundamental commonalities between lexical, morphological and syntactic organisation. Let’s look at a few examples.

Polysemy in the lexicon: over

We begin by considering evidence for polysemy at the level of lexical organisation. The word we will consider is the much studied English preposition *over*. Consider the following examples:

- | | | |
|---------|-----------------------------------|----------------------|
| (19) a. | The picture is over the sofa. | ABOVE |
| b. | The picture is over the hole. | COVERING |
| c. | The ball is over the wall. | ON-THE-OTHER-SIDE-OF |
| d. | The government handed over power. | TRANSFER |
| e. | She has a strange power over me. | CONTROL |

These sentences illustrate various senses of *over*, which are listed in the right-hand column. While each is distinct, they can all be related to one another; they all derive from a central ‘above’ meaning. We will explore this point in more detail later in the book (see Chapter 10).

Polysemy in morphology: agentive -er suffix

Just as words like *over* exhibit polysemy, so do morphological categories. Consider the bound morpheme *-er*, the agentive suffix that was briefly discussed earlier in the chapter:

- (20) a. teacher
b. villager
c. toaster
d. best-seller

In each of the examples in (20), the *-er* suffix adds a slightly different meaning. In (20a) it conveys a human AGENT who regularly or by profession carries out the action designated by the verb, in this instance *teach*. In (20b), *-er* relates to a person who lives in a particular place, here a village. In (20c) *-er* relates to an artefact that has the capacity designated by the verb, here *toast*. In (20d) *-er* relates to a particular quality associated with a type of artefact, here the property of selling successfully. Each of these usages is distinct: a teacher is a person who teaches; a toaster is a machine that performs a toasting function; a best-seller is an artefact like a book that has the property of selling well; and a villager is a person who dwells in a village. Despite these differences, these senses are intuitively related in terms of sharing, to a greater or lesser degree, a defining functional ability or attribute: the ability to teach; the ‘ability’ to toast; the attribute of selling well; and the attribute of dwelling in a specific location. This demonstrates the capacity of morphological categories to exhibit polysemy.

Polysemy in syntax: ditransitive construction

Just as lexical and morphological categories exhibit polysemy, so do syntactic categories. For instance, consider the **ditransitive construction**, discussed by Goldberg (1995). This construction has the following syntax:

- (21) SUBJECT VERB OBJECT 1 OBJECT 2

The ditransitive construction also has a range of conventional abstract meanings associated with it, which Goldberg characterises in the terms shown in (22). Note for the time being that terms like AGENT PATIENT and RECIPIENT are labels for ‘semantic roles’, a topic to which we return in Part III of the book.

- (22) a. SENSE 1: AGENT successfully causes recipient to receive PATIENT
INSTANTIATED BY: verbs that inherently signify acts of giving (e.g. *give, pass, hand, serve, feed*)
e.g. [_{SUBJ} Mary] [_{verb} gave] [_{OBJ 1} John] [_{OBJ 2} the cake]
- b. SENSE 2: conditions of satisfaction imply that AGENT causes recipient to receive PATIENT
INSTANTIATED BY: verbs of giving with associated satisfaction conditions (e.g. *guarantee, promise, owe*)
e.g. Mary promised John the cake
- c. SENSE 3: AGENT causes recipient not to receive PATIENT
INSTANTIATED BY: verbs of refusal (e.g. *refuse, deny*)
e.g. Mary refused John the cake

- d. SENSE 4: AGENT acts to cause recipient to receive PATIENT at some future point in time
INSTANTIATED BY: verbs of future transfer (e.g. *leave, bequeath, allocate, reserve, grant*)
e.g. Mary left John the cake
- e. SENSE 5: AGENT enables recipient to receive PATIENT
INSTANTIATED BY: verbs of permission (e.g. *permit, allow*)
e.g. Mary permitted John the cake
- f. SENSE 6: AGENT intends to cause recipient to receive PATIENT
INSTANTIATED BY: verbs involved in scenes of creation (e.g. *bake, make, build, cook, sew, knit*)
e.g. Mary baked John the cake

While each of the abstract senses associated with ‘ditransitive’ syntax are distinct, they are clearly related: they all concern volitional transfer, although the nature of the transfer, or the conditions associated with the transfer, vary from sense to sense. We will return to discuss constructions like these in more detail in Part III of the book.

In sum, as we saw for categorisation, cognitive linguists argue that polysemy is a phenomenon common to ‘distinct’ areas of language. Both ‘fuzzy’ categories and polysemy, then, are characteristics that unite all areas of human language and thus enable generalisation within the cognitive linguistics framework.

Metaphor

Cognitive linguists also argue that metaphor is a central feature of human language. As we saw in the previous chapter, metaphor is the phenomenon where one conceptual domain is systematically structured in terms of another. One important feature of metaphor is **meaning extension**. That is, metaphor can give rise to new meaning. Cognitive linguists argue that metaphor-based meaning extension can also be identified across a range of ‘distinct’ linguistic phenomena, and that metaphor therefore provides further evidence in favour of generalising across the ‘distinct’ areas of language. In this section we’ll consider lexicon and syntax.

Metaphor in the lexicon: over (again)

In the previous section we observed that the preposition *over* exhibits polysemy. One question that has intrigued cognitive linguists concerns how polysemy is motivated. That is, how does a single lexical item come to have a

multiplicity of distinct yet related meanings associated with it? Lakoff (1987) has argued that an important factor in motivating meaning extension, and hence the existence of polysemy, is metaphor. For instance, he argues that the CONTROL meaning of *over* that we saw in (19e) derives from the ABOVE meaning by virtue of metaphor. This is achieved via application of the metaphor CONTROL IS UP. This metaphor is illustrated by (23):

- (23) a. I'm *on top* of the situation.
 b. She's at the *height* of her powers.
 c. His power *rose*.

These examples illustrate that POWER or CONTROL is being understood in terms of greater elevation (UP). In contrast, lack of power or lack of control is conceptualised in terms of occupying a reduced elevation on the vertical axis (DOWN), as shown by (24):

- (24) a. Her power is on the *decline*.
 b. He is *under* my control.
 c. He's *low* in the company *hierarchy*.

By virtue of the independent metaphor CONTROL IS UP, the lexical item *over*, which has an ABOVE meaning conventionally associated with it, can be understood metaphorically as indicating greater control. Through frequency of use the meaning of CONTROL becomes conventionally associated with *over* in such a way that *over* can be used in non-spatial contexts like (19e), where it acquires the CONTROL meaning.

Metaphor in the syntax: the ditransitive (again)

One of the observations that Goldberg makes in her analysis of the ditransitive construction is that it typically requires a volitional AGENT in subject position. This is because the meaning associated with the construction is one of *intentional* transfer. Unless there is a sentient AGENT who has the capacity for intention, then one entity cannot be transferred to another. However, we do find examples of this construction where the subject (in square brackets) is not a volitional AGENT:

- (25) a. [The rain] gave us some time.
 b. [The missed ball] handed him the victory.

Goldberg argues that examples like these are extensions of the ditransitive construction, and are motivated by the existence of the metaphor CAUSAL EVENTS ARE PHYSICAL TRANSFERS. Evidence for this metaphor comes from examples

like the ones in (26), which illustrate that we typically understand abstract causes in terms of physical transfer:

- (26) a. David Beckham put a lot of swerve on the ball.
b. She gave me a headache.

In these examples causal events like causing a soccer ball to swerve, or causing someone to have a headache, are conceptualised as the transfer of a physical entity. Clearly the English soccer star David Beckham, well known for his ability to ‘bend’ a football around defensive walls, cannot literally put ‘swerve’ on a football; ‘swerve’ is not a physical entity that can be ‘put’ anywhere. However, we have no problem understanding what this sentence means. This is because we ‘recognise’ the convention within our language system of understanding causal events metaphorically in terms of physical transfer.

Goldberg argues that it is due to this metaphor that the ditransitive construction, which normally requires a volitional AGENT, can sometimes have a non-volitional subject like *a missed ball* or *the rain*. The metaphor licenses the extension of the ditransitive so that it can be used with non-volitional AGENTS.

To conclude the discussion so far, this section has illustrated the view held by cognitive linguists that various areas of human language share certain fundamental organising principles. This illustrates the ‘Generalisation Commitment’ adopted by cognitive linguists. One area in which this approach has achieved considerable success is in uniting the lexical system with the grammatical system, providing a unified theory of grammatical and lexical structure. As we will see in Part III, cognitive approaches to grammar treat lexicon and syntax not as distinct components of language, but instead as a continuum. However, the relationship between phonology and other areas of human language has only recently begun to be explored from a cognitive perspective. For this reason, while aspects of the foregoing discussion serve to illustrate some similarities between the phonological subsystem and the other areas of the language system, we will have relatively little to say about phonology in the remainder of this book.

2.1.2 The ‘Cognitive Commitment’

We turn next to the ‘Cognitive Commitment’. We saw above that the ‘Generalisation Commitment’ leads to the search for principles of language structure that hold across all aspects of language. In a related manner, the ‘Cognitive Commitment’ represents the view that principles of linguistic structure should reflect what is known about human cognition from other disciplines, particularly the other cognitive sciences (philosophy, psychology,

artificial intelligence and neuroscience). In other words, it follows from the ‘Cognitive Commitment’ that language and linguistic organisation should reflect general cognitive principles rather than cognitive principles that are specific to language. Accordingly, cognitive linguistics rejects the **modular theory** of mind that we mentioned above (section 2.1.1). The modular theory of mind is associated particularly with formal linguistics, but is also explored in other areas of cognitive science such as philosophy and cognitive psychology, and holds that the human mind is organised into distinct ‘encapsulated’ modules of knowledge, one of which is language, and that these modules serve to ‘digest’ raw sensory input in such a way that it can then be processed by the central cognitive system (involving deduction, reasoning, memory and so on). Cognitive linguists specifically reject the claim that there is a distinct **language module**, which asserts that linguistic structure and organisation are markedly distinct from other aspects of cognition (see Chapter 4). Below we consider three lines of evidence that, according to cognitive linguists, substantiate the view that linguistic organisation reflects more general cognitive function.

Attention: profiling in language

A very general cognitive ability that human beings have is **attention**, together with the ability to shift attention from one aspect of a scene to another. For instance, when watching a tennis match we can variously attend to the umpire, the flight of the ball back and forth, one or both of the players or parts of the crowd, zooming ‘in and out’ so to speak. Similarly, language provides ways of directing attention to certain aspects of the scene being linguistically encoded. This general ability, manifest in language, is called **profiling** (Langacker 1987, among others; see also Talmy’s (2000) related notion of attentional **windowing**).

One important way in which language exhibits profiling is in the range of grammatical constructions it has at its disposal, each of which serves to profile different aspects of a given scene. For instance, given a scene in which a boy kicks over a vase causing it to smash, different aspects of the scene can be linguistically profiled:

- (27) a. The boy kicks over the vase.
- b. The vase is kicked over.
- c. The vase smashes into bits.
- d. The vase is in bits.

In order to discuss the differences between the examples in (27), we’ll be relying on some grammatical terminology that may be new to the reader. We will explain these terms briefly as we go along, but grammatical terms are explained in more detail in the grammar tutorial in Chapter 14.

The aspects of the scene profiled by each of these sentences are represented in Figure 2.2. Figure 2.2(a) corresponds to sentence (27a). This is an active sentence in which a relationship holds between the initiator of the action (the boy) and the object that undergoes the action (the vase). In other words, the boy is the AGENT and the vase is the PATIENT. In Figure 2.2(a) both AGENT and PATIENT are represented by circles. The arrow from the AGENT to the PATIENT represents the transfer of energy, reflecting the fact that the AGENT is acting upon the PATIENT. Moreover, both AGENT and PATIENT, as well as the energy transfer, are represented in bold. This captures the fact that the entire **action chain** is being profiled, which is the purpose of the active construction.

Now let's compare sentence (27b). This is a passive sentence, and is represented by Figure 2.2(b). Here, the energy transfer and the PATIENT are being profiled. However, while the AGENT is not mentioned in the sentence, and hence is not in profile, it must be understood as part of the background. After all, an action chain requires an AGENT to instigate the transfer of energy. To represent this fact, the AGENT is included in Figure 2.2(a), but is not featured in bold, reflecting the position that the AGENT is contextually understood but not in profile.

The third sentence, example (27c), profiles the change in the state of the vase: the fact that it smashes into bits. This is achieved via a subject-verb-complement construction. A complement is an obligatory element that is required by another element in a sentence to complete its meaning. In (27c), the complement is the expression *into bits*, which completes the meaning of the expression *smashes*. This is captured by Figure 2.2(c). In figure 2.2(c) it is the internal change of state of the vase that is profiled. The arrow within the circle (the circle depicts the vase) shows that the vase is undergoing an internal change of state. The state the vase is 'moving to' is represented by the box with the letter 'b' inside it. This stands for the state IN BITS. In this diagram the entity, the change of state and the resulting state are all in bold, reflecting the fact that all these aspects of the action chain are being profiled by the corresponding sentence.

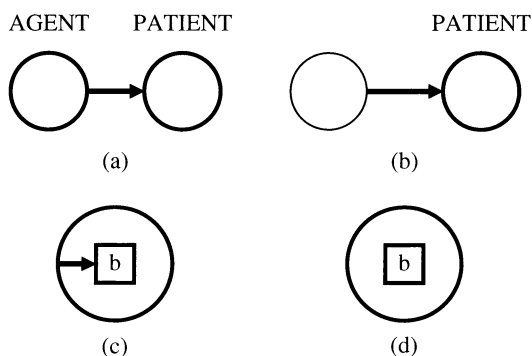


Figure 2.2 Profiling

Finally, consider sentence (27d). The grammatical form of this sentence is the subject-copula-complement construction. The copula is the verb *be*, which is specialised for encoding a particular state. In this case the state is IN BITS, which is captured in Figure 2.2(d).

In sum, each of the constructions ACTIVE, PASSIVE, SUBJECT-VERB-COMPLEMENT and SUBJECT-COPULA-COMPLEMENT is specialised for profiling a particular aspect of an action chain. In this way, linguistic structure reflects our ability to attend to distinct aspects of a scene. These examples demonstrate how linguistic organisation reflects a more general cognitive ability: attention.

It is worth observing at this point that constructions of the kind we have just discussed are not restricted to encoding a canonical action chain (one involving the transfer of energy). For example, the active construction can often be applied in cases where an action is not involved. Consider **stative verbs**, like *own*. A stative verb encodes a relatively stable state that persists over time. This verb can appear in active or passive constructions, even though it describes a state rather than an action:

- (28) a. John not Steve owns the shop on Trafalgar Street. [active]
 b. The shop on Trafalgar Street is owned [passive]
 by John not Steve.

In Part III of the book, we will return in more detail to the issue of grammatical constructions and the range of meanings associated with them.

Categorisation: fuzzy categories

We saw above that entities like cups constitute fuzzy categories, which are characterised by the fact that they contain members that are more or less representative of the category. This results in a set of members related by family resemblance rather than a single criterial feature, or a limited set of criterial features possessed by every member of the category. In other words, categories formed by the human mind are rarely 'neat and tidy'. We also saw that fuzzy categories are a feature of language in that members of linguistic categories, despite important similarities, often show quite distinct behaviour. In other words, according to the cognitive framework, the same principles that hold for categorisation in general also hold for linguistic categorisation.

Metaphor

As we began to see in the previous chapter, and as we will see in further detail in Chapter 9, the view adopted in cognitive linguistics is that metaphor is

a conceptual rather than a purely linguistic phenomenon. Moreover, the key proponents of the conceptual metaphor approach, George Lakoff and Mark Johnson (1980, 1999), argue that many of the ways in which we think and act are fundamentally metaphorical in nature.

For instance, we conceptualise institutions like governments, universities, and businesses in terms of a hierarchy. Diagrams of such institutions place the person with the highest rank at the top or ‘head’, while the person with the lowest rank is placed at the lowest point or ‘bottom’. In other words, hierarchies are conceptualised and represented non-linguistically in terms of the conceptual metaphor CONTROL/POWER IS UP.

Just as metaphors like CONTROL IS UP show up in a range of modalities, that is different ‘dimensions’ of expression such as social organisation, pictorial representation or gesture, among others, we have begun to see that they are also manifest in language. The English preposition *over* has a conventional CONTROL meaning associated with it, precisely because of meaning extension due to the conceptual metaphor CONTROL IS UP.

In the foregoing discussion, we have explored three ways in which aspects of general cognition show up in language. Evidence of this kind forms the basis of the cognitive argument that language reflects general cognition.

2.2 The embodied mind

In this section, we turn to **embodiment**, a central idea in cognitive linguistics. Since the seventeenth-century French philosopher René Descartes developed the view that mind and body are distinct entities – the principle of **mind/body dualism** – there has been a common assumption within philosophy and the other more recent cognitive sciences that the mind can be studied without recourse to the body, and hence without recourse to embodiment. In modern linguistics this **rationalist approach** has been most evident in formal approaches such as the **Generative Grammar** approach developed by Noam Chomsky (see Chapter 22) and formal approaches to semantics, such as the framework developed by Richard Montague (see Chapter 13). Proponents of these approaches argue that it is possible to study language as a formal or computational system, without taking into account the nature of human bodies or human experience.

In contrast, cognitive linguistics is not rationalist in this sense, but instead takes its inspiration from traditions in psychology and philosophy that emphasise the importance of human experience, the centrality of the human body, and human-specific cognitive structure and organisation, all of which affect the nature of our experience. According to this **empiricist view**, the human mind – and therefore language – cannot be investigated in isolation from human embodiment.

2.2.1 Embodied experience

The idea that experience is embodied entails that we have a species-specific view of the world due to the unique nature of our physical bodies. In other words, our construal of reality is likely to be mediated in large measure by the nature of our bodies.

One obvious way in which our embodiment affects the nature of experience is in the realm of colour. While the human visual system has three kinds of photo-receptors or colour channels, other organisms often have a different number. For instance, the visual system of squirrels, rabbits and possibly cats, makes use of two colour channels, while other organisms, like goldfish and pigeons, have four colour channels. Having a different range of colour channels affects our experience of colour in terms of the range of colours accessible to us along the colour spectrum. Some organisms can see in the infrared range, like rattlesnakes, which hunt prey at night and can visually detect the heat given off by other organisms. Humans are unable to see in this range. As this simple example demonstrates, the nature of our visual apparatus – one aspect of our physical embodiment – determines the nature and range of our visual experience.

Similarly, the nature of our biological morphology (the kinds of body parts we have), together with the nature of the physical environment with which we interact, determines other aspects of our experience. For instance, while gravity is an objective feature of the world, our experience of gravity is determined by our bodies and by the ecological niche we inhabit. For instance, hummingbirds – which can flap their wings up to a remarkable fifty times per second – respond to gravity in a very different way from humans. In order to overcome gravity, hummingbirds are able to rise directly into the air without pushing off from the ground, due to the rapid movement of their wings. Moreover, due to their small size, their experience of motion is rather different from ours: hummingbirds can stop almost instantaneously, experiencing little momentum. Compare this with the experience of a sprinter at the end of a 100m race: a human cannot stop instantaneously but must take a few paces to come to a standstill.

Now consider organisms that experience gravity in an even more different way. Fish, for example, experience very little gravity, because water reduces its effect. This explains their morphology, which is adapted to the ecological niche they inhabit and enables motion through a reduced-gravity environment. The neuroscientist Ernst Pöppel (1994) has even suggested that different organisms might have different kinds of neural ‘timing mechanisms’ which underpin abilities such as event perception (see Chapter 3). This is likely to affect their experience of time. The idea that different organisms have different kinds of experiences due to the nature of their embodiment is known as **variable embodiment**.

2.2.2 Embodied cognition

The fact that our experience is embodied – that is, structured in part by the nature of the bodies we have and by our neurological organisation – has consequences for cognition. In other words, the concepts we have access to and the nature of the ‘reality’ we think and talk about are a function of our embodiment: we can only talk about what we can perceive and conceive, and the things that we can perceive and conceive derive from embodied experience. From this point of view, the human mind must bear the imprint of embodied experience.

In his now classic 1987 book, *The Body in the Mind*, Mark Johnson proposes that one way in which embodied experience manifests itself at the cognitive level is in terms of **image schemas** (see Chapter 6). These are rudimentary concepts like CONTACT, CONTAINER and BALANCE, which are meaningful because they derive from and are linked to human **pre-conceptual experience**: experience of the world directly mediated and structured by the human body. These image-schematic concepts are not disembodied abstractions, but derive their substance, in large measure, from the sensory-perceptual experiences that give rise to them in the first place. Lakoff (1987, 1990, 1993) and Johnson (1987) have argued that embodied concepts of this kind can be systematically extended to provide more abstract concepts and conceptual domains with structure. This process is called **conceptual projection**. For example, they argue that conceptual metaphor (which we discussed briefly above and to which we return in detail in Chapter 9) is a form of conceptual projection. According to this view, the reason we can talk about being *in* states like love or trouble (29) is because abstract concepts like LOVE are structured and therefore understood by virtue of the fundamental concept CONTAINER. In this way, embodied experience serves to structure more complex concepts and ideas.

- (29) a. George is in love.
b. Lily is in trouble.
c. The government is in a deep crisis.

The developmental psychologist Jean Mandler (e.g. 1992, 1996, 2004) has made a number of proposals concerning how image schemas might arise from embodied experience. Starting at an early age, and certainly by two months, infants attend to objects and spatial displays in their environment. Mandler suggests that by attending closely to such spatial experiences, children are able to abstract across similar kinds of experiences, finding meaningful patterns in the process. For instance, the CONTAINER image schema is more than simply a spatio-geometric representation. It is a ‘theory’ about a particular kind of configuration in which one entity is supported by another entity that contains it. In other

words, the CONTAINER schema is meaningful because containers are meaningful in our everyday experience. Consider the spatial scene described in (30).

(30) The coffee is in the cup.

Tyler and Evans make the following observations about this spatial scene:

... the spatial scene relating to *in* involves a containment function, which encompasses several consequences such as locating and limiting the activities of the contained entity. Being contained in the cup prevents the coffee from spreading out over the table; if we move the cup, the coffee moves with it. (Tyler and Evans 2003: ix)

It is for this reason that the English preposition *in* can be used in scenes that are non-spatial in nature, like the examples in (29). It is precisely because containers constrain activity that it makes sense to conceptualise POWER and all-encompassing states like LOVE or CRISIS in terms of CONTAINMENT. Mandler (2004) describes this process of forming image schemas in terms of a redescription of spatial experience via a process she labels **perceptual meaning analysis**. As she puts it, '[O]ne of the foundations of the conceptualizing capacity is the image schema, in which spatial structure is mapped into conceptual structure' (Mandler 1992: 591). She further suggests that 'Basic, recurrent experiences with the world form the bedrock of the child's semantic architecture, which is already established well before the child begins producing language' (Mandler 1992: 597). In other words, it is experience, meaningful to us by virtue of our embodiment, that forms the basis of many of our most fundamental concepts.

2.2.3 Experiential realism

An important consequence of viewing experience and conceptualisation as embodied is that this affects our view of what reality is. A widely held view in formal semantics is that the role of language is to describe states of affairs in the world. This rests on the assumption that there is an objective world 'out there', which language simply reflects. However, cognitive linguists argue that this **objectivist approach** misses the point that there cannot be an objective reality that language reflects directly, because reality is not objectively given. Instead, reality is in large part constructed by the nature of our unique human embodiment. This is not to say that cognitive linguists deny the existence of an objective physical world independent of human beings. After all, gravity exists, and there is a colour spectrum (resulting from light striking surfaces of different kinds and densities), and some entities give off heat, including body

heat, which can only be visually detected in the infrared range. However, the parts of this external reality to which we have access are largely constrained by the ecological niche we have adapted to and the nature of our embodiment. In other words, language does not directly reflect the world. Rather, it reflects our unique human construal of the world: our ‘world view’ as it appears to us through the lens of our embodiment. In Chapter 1 we referred to human reality as ‘projected reality’, a term coined by the linguist Ray Jackendoff (1983).

This view of reality has been termed **experientialism** or **experiential realism** by cognitive linguists George Lakoff and Mark Johnson. Experiential realism assumes that there is a reality ‘out there’. Indeed, the very purpose of our perceptual and cognitive mechanisms is to provide a representation of this reality, and thus to facilitate our survival as a species. After all, if we were unable to navigate our way around the environment we inhabit and avoid dangerous locations like clifftops and dangerous animals like wild tigers, our cognitive mechanisms would be of little use to us. However, by virtue of being adapted to a particular ecological niche and having a particular form and configuration, our bodies and brains necessarily provide one particular perspective among many possible and equally viable perspectives. Hence, experiential realism acknowledges that there is an external reality that is reflected by concepts and by language. However, this reality is mediated by our uniquely human experience which constrains the nature of this reality ‘for us’.

2.3 Cognitive semantics and cognitive approaches to grammar

Having set out some of the fundamental assumptions behind the cognitive approach to language, in this section we briefly map out the field of cognitive linguistics. Cognitive linguistics can be broadly divided into two main areas: **cognitive semantics** and **cognitive (approaches to) grammar**. However, unlike formal approaches to linguistics, which often emphasise the role of grammar, cognitive linguistics emphasises the role of meaning. According to the cognitive view, a model of meaning (a cognitive semantics) has to be delineated before an adequate cognitive model of grammar can be developed. Hence a cognitive grammar assumes a cognitive semantics and is dependent upon it. This is because grammar is viewed within the cognitive framework as a meaningful system in and of itself, which therefore shares important properties with the system of linguistic meaning and cannot be meaningfully separated from it.

The area of study known as cognitive semantics, which is explored in detail in Part II of the book, is concerned with investigating the relationship between experience, the conceptual system and the semantic structure encoded by language. In specific terms, scholars working in cognitive semantics investigate knowledge representation (**conceptual structure**) and meaning construction (**conceptualisation**). Cognitive semanticists have employed language as the

lens through which these cognitive phenomena can be investigated. It follows that cognitive semantics is as much a model of mind as it is a model of linguistic meaning.

Cognitive grammarians have also typically adopted one of two foci. Scholars like Ronald Langacker have emphasised the study of the cognitive principles that give rise to linguistic organisation. In his theoretical framework **Cognitive Grammar**, Langacker has attempted to delineate the principles that serve to structure a grammar, and to relate these to aspects of general cognition. Because the term 'Cognitive Grammar' is the name of a specific theory, we use the (rather cumbersome) expression 'cognitive (approaches to) grammar' as the general term for cognitively oriented models of the language system.

The second avenue of investigation, pursued by researchers including Fillmore and Kay (Fillmore *et al.* 1988; Kay and Fillmore 1999), Lakoff (1987), Goldberg (1995) and more recently Bergen and Chang (2005) and Croft (2002), aims to provide a more descriptively detailed account of the units that comprise a particular language. These researchers have attempted to provide an inventory of the units of language. Cognitive grammarians who have pursued this line of investigation are developing a collection of theories that can collectively be called **construction grammars**. This approach takes its name from the view in cognitive linguistics that the basic unit of language is a form-meaning symbolic assembly which, as we saw in Chapter 1, is called a **construction**.

It follows that cognitive approaches to grammar are not restricted to investigating aspects of grammatical structure largely independently of meaning, as is often the case in formal traditions. Instead, cognitive approaches to grammar encompass the entire inventory of linguistic units defined as form-meaning pairings. These run the gamut from skeletal syntactic configurations like the ditransitive construction we considered earlier, to idioms, to bound morphemes like the *-er* suffix, to words. This entails that the received view of clearly distinct 'sub-modules' of language cannot be meaningfully upheld within cognitive linguistics, where the boundary between cognitive semantics and cognitive (approaches to) grammar is less clearly defined. Instead, meaning and grammar are seen as two sides of the same coin: to take a cognitive approach to grammar is to study the units of language and hence the language system itself. To take a cognitive approach to semantics is to attempt to understand how this linguistic system relates to the conceptual system, which in turn relates to embodied experience. The concerns of cognitive semantics and cognitive (approaches to) grammar are thus complementary. This idea is represented in Figure 2.3. The organisation of this book reflects the fact that it is practical to divide up the study of cognitive linguistics into these two areas for purposes of teaching and learning. However, this should not be taken as an indication that these two areas of cognitive linguistics are independent areas of study or research.

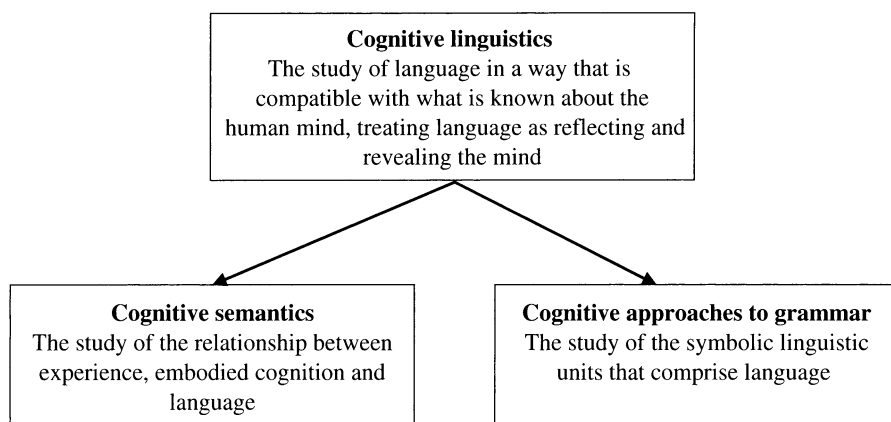


Figure 2.3 The study of meaning and grammar in cognitive linguistics

2.4 Summary

In this chapter, we have provided an overview of the assumptions and commitments that make cognitive linguistics a distinctive enterprise. We have outlined two key commitments widely shared by cognitive linguists. These are the ‘**Generalisation Commitment**’ and the ‘**Cognitive Commitment**’. These two commitments underlie the orientation and approach adopted by cognitive linguists, and the assumptions and methodologies employed in the two main branches of the cognitive linguistics enterprise, **cognitive semantics** and **cognitive (approaches to) grammar**. We also introduced the **embodied cognition** thesis which is central to much research in cognitive linguistics and addresses the nature of the relationship between language, mind and experience. The view taken in cognitive linguistics is that conceptual organisation within the human mind is a function of the way our species-specific bodies interact with the environment we inhabit. Finally, we provided a brief overview of cognitive semantics and cognitive approaches to grammar which are addressed in detail in Part II and Part III of the book, respectively.

Further reading

Assumptions in cognitive linguistics

The following are all articles by leading cognitive linguists that set out the assumptions and the nature of the cognitive linguistics enterprise:

- **Fauconnier (1999)**. A discussion of methodological issues and the nature of the approach adopted in cognitive linguistics, particularly with respect to meaning. Fauconnier, one of the early pioneers in

cognitive linguistics, illustrates with examples from the theory of conceptual blending, which he developed in joint work with Mark Turner.

- **Lakoff (1990)**. In the first part of this important article, published in the very first volume of the journal *Cognitive Linguistics*, Lakoff discusses issues relating to the ‘Generalisation Commitment’ and the ‘Cognitive Commitment’. He also explains how cognitive linguistics differs from Generative Grammar.
- **Langacker (1999a)**. An important article by another pioneering figure in cognitive linguistics. In this article, Langacker evaluates the approach and methodologies employed in cognitive linguistics and relates this to the formalist and functionalist traditions in linguistics. He illustrates with a discussion from some of the key constructs in his Cognitive Grammar framework.
- **Talmy (2000: Vol. I, 1–18)**. In the introduction to his two-volume edifice, *Toward a Cognitive Semantics*, Talmy outlines his view of the cognitive linguistics enterprise and describes how his own work fits in with and has contributed to this endeavour.

Embodied cognition

- **Clark (1997)**. Drawing on recent work in robotics, neuroscience, psychology and artificial intelligence, Clark, a leading cognitive scientist, presents a compelling and highly accessible overview of the new science of the embodied mind.
- **Evans (2004a)**. This book addresses how time, a fundamental aspect of human experience, is conceptualised. The discussion relates neurological, phenomenological and sensory-perceptual aspects of embodied experience to the experience of temporal cognition as revealed by language. Chapter 4 provides a presentation of some key arguments for the cognitive linguistics perspective on embodied cognition.
- **Lakoff (1987)**. This is a classic work by one of the pioneers in cognitive linguistics. Part II of the book is particularly important for the development of experiential realism.
- **Lakoff and Johnson (1980)**. This short volume laid the foundations for the approach to embodied cognition in cognitive linguistics.
- **Lakoff and Johnson (1999)**. This represents an updated account of experiential realism as developed by Lakoff and Johnson (1980).
- **Mandler (2004)**. Influential developmental psychologist Jean Mandler argues for the role of image schemas in the development of conceptual structure and organisation.

- **Varela, Thompson and Rosch (1991).** A highly influential book on embodiment, cognition and human experience by leading cognitive scientists.

Exercises

2.1 Categorisation and family resemblance

The philosopher Wittgenstein famously argued that the category *GAME* exhibits family resemblance. To test this, first make a list of as many different kinds of games as you can think of. Now see if there is a limited set of conditions that is common to this entire list ('necessary' conditions) and sufficient to distinguish this category from other related categories ('sufficient' conditions) like competitions, amusement activities and so on. Do your conclusions support or refute Wittgenstein's claim?

Now see if you can identify the ways in which the different games you list share family resemblance 'traits'. Try to construct a 'radial' network showing the degrees of family resemblance holding between games of different kinds. A radial network is a diagram in which the most/more prototypical game(s) is/are placed at the centre and less prototypical games are less central, radiating out from the centre.

2.2 Polysemy

Consider the word *head*. Try and come up with as many different meanings for this word as possible. You may find it helpful to collect or create sentences involving the word.

Now consider the closed-class word *you*. Cognitive linguists assume that even closed-class words exhibit polysemy. Collect as many sentences as you can involving *you* and try and identify differences in how this word is used. Do your findings support the view that this word exhibits polysemy?

2.3 Metaphor

Reconsider the different meanings for *head* that you uncovered in the previous exercise. Would you class any of these distinct meanings as metaphorical? Explain your reasoning. Now try and give an account of what motivated the extension from the 'core' meaning of *head* to the metaphoric usage(s).

2.4 Image schemas

The spatial meanings associated with prepositions present a clear case of the way in which image schemas underpin language. In view of this, what sets of

image schemas might underpin the semantic distinction between the prepositions *up/down* and *above/under*?

Now consider the metaphoric use of the prepositions *on* and *in* in the following sentences:

- (a) The guard is on duty.
- (a') The shoes are on sale.
- (b) Munch's painting *The Scream* portrays a figure in despair.
- (b') Sven is in trouble with Nancy.

What might be the experiential basis for the fact that states like SALES and DUTY are described in terms of ON, while states like DESPAIR and TROUBLE are described in terms of IN? We saw in this chapter that the CONTAINER image schema plausibly underpins IN. What might be the image schema underpinning ON?

Universals and variation in language, thought and experience

As we saw in Chapter 2, the cognitive linguistics enterprise is characterised by two commitments: (1) the ‘**Generalisation Commitment**’ – a commitment to the characterisation of general principles that are responsible for all aspects of human language; and (2) the ‘**Cognitive Commitment**’ – a commitment to providing a characterisation of general principles for language that accords with what is known about the mind and brain from other disciplines (Lakoff 1990). An important consequence of this approach is the position that language does not result from an encapsulated ‘module’ of specialised knowledge, separable from general cognition (in contrast with the view developed in formal approaches to linguistics), but instead that language reflects and is informed by non-linguistic aspects of cognition. In particular, given the premise that the principles that inform language reflect general cognitive principles, the language system itself can be seen as a window that enables the direct investigation of **conceptual structure** (knowledge representation, including the structure and organisation of concepts) and **conceptualisation** (the process of meaning construction).

Although cognitive linguists have often been concerned with investigating the general cognitive principles (common to all humans) that govern language, it does not follow from this that all languages are the same, either in terms of grammatical structure or semantic structure. In this chapter, we review some influential cognitively oriented studies that demonstrate that languages can exhibit radically different conceptual organisation and structure. It seems that common cognitive principles do not give rise to uniform linguistic organisation and structure. On the contrary, cross-linguistic variation is widespread. At the same time, the existence of certain common patterns across languages is a matter of empirical fact. These common patterns are known as **linguistic**

universals. For cognitive linguists, these commonalities are explained by the existence of general cognitive principles shared by all humans, in addition to the fundamentally similar experiences of the world also shared by all humans due to embodiment. Nevertheless, given the premise that language reflects cognitive organisation, the existence of cross-linguistic variation entails that speakers of different languages have different underlying conceptual systems. This view has implications for the thesis of **linguistic relativity** or **linguistic determinism** – the view that the language you speak affects or determines how you see the world, most famously expounded in the writings of Benjamin Lee Whorf in the 1930s and 1940s. Hence, once we have developed the cognitive linguistics approach to linguistic universals and cross-linguistic variation as we see it, we will re-examine the Whorfian linguistic relativity principle.

3.1 Universals in thought and language

We begin by considering the issue of linguistic universals. It is important to observe here that the term ‘linguistic universal’ can be understood in two quite distinct ways. On the one hand, the term can refer to patterns of similarity that are attested in typological studies: these are usually large-scale comparative studies that set out to discover linguistic patterns in relation to a given phenomenon. The existence of the typological universals uncovered by these studies is a matter of empirical fact and is uncontroversial. On the other hand, the term ‘universal’ can also be used to refer to underlying principles of linguistic organisation and structure that are represented in the human mind. This view is most prominently associated with the generative grammar framework developed by Noam Chomsky, which assumes the existence of a **Universal Grammar**: a set of innate universal principles that equips all humans to acquire their native language and is also held to account for patterns of cross-linguistic similarity. This view is controversial for many linguists, including cognitive linguists. We will briefly set out the assumptions of the Generative Grammar model below (section 3.1.2), and return to these issues in more detail towards the end of the book (Chapter 22), but consider for the time being the following extract from Levinson (1996):

It may be claimed, the Kantian categories of space, time, cause and so on, form the fundamental ground of our reasoning; they cannot be inferred from experience, but are what we bring to the interpretation of experience from our biological endowment. Thus the conceptual architecture, the essential conceptual parameters, are, as Leibniz would have it, ‘innate ideas’. This line of thought dominates current speculations in the cognitive sciences. It is a view reinforced from many quarters: evolutionary biology and neurophysiology stress the

closeness of our neurological equipment to that of our mammalian cousins, studies of human development (following Piaget) assume an unfolding of inborn potential, psychological models of processing are often presumed to be models of 'hardware' properties rather than models of learned or acquired tendencies or 'software', and so on. In linguistics, the adoption of natural science ideals has led to the search for universals without parallel concern for language differences. (Levinson 1996: 133)

As Levinson's comment suggests, the search for linguistic universals (in the sense of universal cognitive principles of language) has preoccupied much of modern linguistics, particularly since the advent of Chomsky's work on generative grammar in the 1950s. However, as Levinson observes, the search for Universal Grammar has prompted some linguists to argue that quite radical cross-linguistic variation has been ignored by formal linguists. To provide just a few examples, languages can range from having between eleven and 141 distinctive speech sounds; some languages lack morphological marking for properties like number (singular or plural) or tense; and some languages appear to lack syntactic constraints on word order, or fail to exhibit familiar word classes such as adjective.

Despite the widespread view within formal linguistics that linguistic structure across languages is broadly similar (and can eventually be stated in terms of a small set of universal principles known as Universal Grammar), studies set within this tradition tend not to be concerned with large-scale cross-linguistic comparison. The branch of linguistics that is concerned with large-scale cross-linguistic comparison, **linguistic typology**, reveals the relative rarity of absolute universals in the sense of patterns of similarity that hold across all languages. Instead, the universals that do emerge are conditional generalisations that can be established to have some statistical validity, as we will see below (section 3.1.1).

As we have already noted, cognitive linguists assume that language reflects conceptual structure and organisation. It follows from this assumption that cross-linguistic differences should point to underlying conceptual differences. Cognitive linguists therefore argue that evidence of variation across languages suggests that languages encode very different kinds of conceptual systems. However, these distinct conceptual systems are thought to emerge from a common **conceptualising capacity**, which derives from fundamental shared aspects of human cognition. Rather than positing universal linguistic principles, then, cognitive linguists posit a common set of cognitive abilities, which serve to both facilitate and constrain the development of our **conceptual systems** (our repository of concepts). Although cross-linguistic analysis reveals that the range of possible conceptual systems found in language is

delimited in certain fundamental ways, the languages of the world can and do exhibit a wide range of variation. Cognitive linguists argue that this fact, revealed by typologists, seriously undermines the position that there can be universal principles of language of the kind posited by formal linguists.

3.1.1 Typological universals

According to Croft (2003: 1–2), the term ‘linguistic typology’ is used in three distinct ways to refer to three different types of approach that fall within the broader discipline of linguistic typology. The first approach, which he calls **typological classification**, involves the assignment of a given language to a single type, based on its properties in a certain area (morphology, word order and so on). The nineteenth- and early twentieth-century typological approach is a representative example, where the emphasis was on developing descriptive taxonomies. For example, in traditional morphological classification, a language is classified as belonging to the ‘isolating’ type if it lacks grammatical affixes, while a language is classified as belonging to the ‘agglutinating’ type if it has grammatical affixes that each encode a single grammatical feature.

The second approach within linguistic typology is what Croft calls **typological generalisation**. This involves the search for systematic patterns across languages (linguistic universals), and identifies what patterns of variation can be predicted to exist on the basis of those observed patterns. This approach has its roots in the work begun by Joseph Greenberg in the 1960s, and in emphasising the predictions that emerge from attested patterns about what is a possible human language goes a step further than the essentially taxonomic approach of typological classification.

The third approach within linguistic typology is what Croft calls **functional typology**. This modern approach rests upon typological generalisation, but goes a step further in developing a theoretical framework that seeks to set out explanations for the observed patterns. This approach is called ‘functional’ typology because it explains these patterns in terms of how language is used for purposes of communication. Functional typology has been developed by typologists such as Bernard Comrie, Talmy Givón, John Haiman, Paul Hopper and William Croft, among others.

Modern linguistic typology adopts large-scale cross-linguistic sampling as its methodology. The size of the sample varies according to the extent to which the phenomenon under investigation is widespread, as well as being constrained by practical considerations; the typical sample size is in the region of 100–200 languages (out of the estimated six thousand living languages in the world). It is important that the generalisations stated by typologists have statistical validity, otherwise they cannot be upheld. The languages that make up these samples are carefully selected, taking into consideration factors that

might affect the reliability of the resulting generalisations, such as genetic relationships between languages and contact between neighbouring but genetically unrelated languages.

Linguistic typologists have discovered that, although it is possible to state certain properties that hold for all languages (unrestricted universals), cross-linguistic variation is ubiquitous. However, typologists have also discovered that, while languages can and do vary, cross-linguistic variation is **constrained**, and these constraints can be stated in terms of implicational universals. Indeed, from the perspective of linguistic typology, it is the constraints on variation that make up the universals of language, rather than a set of universal principles that capture the properties that languages have in common (Universal Grammar). Let's look more closely at the distinction between unrestricted universals and implicational universals, which makes this point clearer.

An **unrestricted universal** states that all languages show a particular pattern with respect to some structural feature, while the other logically possible pattern(s) are unattested. Croft (2003: 52) provides the example in (1).

- (1) All languages have oral vowels.

This means that the other logical possibility, that there are languages without oral vowels, is not attested. This type of unrestricted universal pinpoints cross-linguistic similarity and is relatively uninteresting to typologists because it does not reveal a pattern in the same way that cross-linguistic differences do.

It is much more common for typologists to state **implicational universals**, which do not state that all languages show the same pattern with respect to a given phenomenon, but instead state the restrictions on the logically possible patterns, usually in the following format: 'If language X has property Y, then it will also have property Z'. As Croft (2003: 54) points out, this type of universal pinpoints patterns in variation rather than similarity, since each implicational universal sets out a set of distinct attested possibilities. Croft provides the example in (2), which was proposed by Hawkins (1983: 84, cited in Croft 2003: 53). This implicational universal rests upon the four logically possible patterns listed in (3).

- (2) If a language has noun before demonstrative, then it has noun before relative clause.
- (3) a. languages where both demonstratives and relative clauses follow the noun
b. languages where both demonstratives and relative clauses precede the noun

- c. languages where demonstrative precedes and relative clause follows the noun
- d. languages where demonstrative follows and relative clause precedes the noun

Observe that the implicational universal in (2) excludes the possibility described in (3d). In this way, the implicational universal states the limits on cross-linguistic variation by restricting the possibilities to those described in (3a)–(3c), and entails an absolute universal by stating that the pattern in (3d) is unattested. In reality, most of the universals posited by typologists are of this kind, or indeed of a more complex kind. Croft describes the differences between typological and generative approaches as follows:

One of the major differences between the generative and typological approaches is what direction to generalize first. Given a grammatical phenomenon such as a relative clause structure in English, one could generalize in several directions. One could compare the relative clause structure with other complex sentence structures in English . . . and then generalize over these different structures in English. This is the classic structuralist-generative approach. Alternatively, one could compare relative clause structure in English with relative clause structure in other languages, and then generalize over relative clauses in human languages. This is the classic typological approach . . . the typologist begins with cross-linguistic comparisons, and then compares typological classifications of different structural phenomena, searching for relationships. In contrast, the generative linguist begins with language-internal structural generalizations and searches for correlations of internal structural facts, and only then proceeds to cross-linguistic comparison. (Croft 2003: 285)

A further important difference between functional typology and the generative approach is that functional typologists reject the idea of specialised innate linguistic knowledge (Universal Grammar). Instead, functional typology comes much closer to cognitive linguistics in orientation, in two important ways. Firstly, functional typology emphasises language function and use in developing explanations for linguistic phenomena. Secondly, functional typology appeals to non-linguistic aspects of cognition to explain the properties of language. For example, many typologists adopt some version of a **semantic map model** in accounting for typological patterns (Croft 2003: 133). A semantic map is a language-specific typological pattern, which rests upon a universal **conceptual space** or system of knowledge. We return to look at this idea in more detail at the end of Part III (Chapter 20).